Structuring and multi-modal management of a corpus of image for the spatio-temporal monitoring of a heritage restoration worksite

Laura Willot[1,2,3],

Livio De Luca[1], Valérie Gouet-Brunet[2], Dan Vodislav[3], Adeline Manuel[1]

[1] MAP UPR 2002 CNRS, Marseille, France
[2] LASTIG, Univ. Gustave Eiffel, ENSG, IGN, 94160 Saint-Mande, France
[3] ETIS Laboratory UMR 8051, CY Cergy Paris Université, ENSEA, CNRS, France

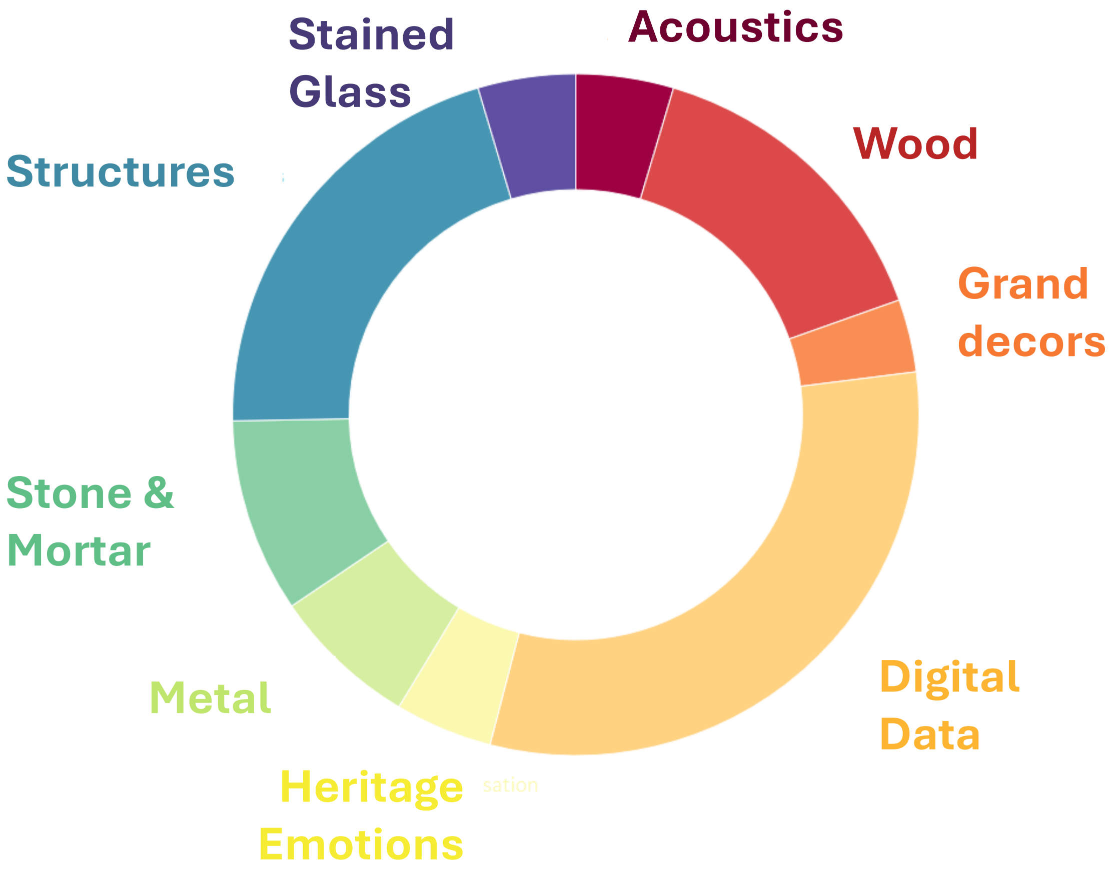
Scientific Context

- Notre-Dame de Paris, “Le Chantier du siècle”
  - Cultural Heritage building
  - 15th April 2019, the fire
  - 2019, the heritage restoration worksite

The Notre-Dame cathedral is ravaged by fire, 15th April 2019, Paris. (Bruno de Hogues / Only France / AFP)

Scientific Context

- Notre-Dame de Paris, the Scientific mobilisation
  - 9 thematic Working Groups (WG)
  - Study, analyse the building and accompany the restoration work

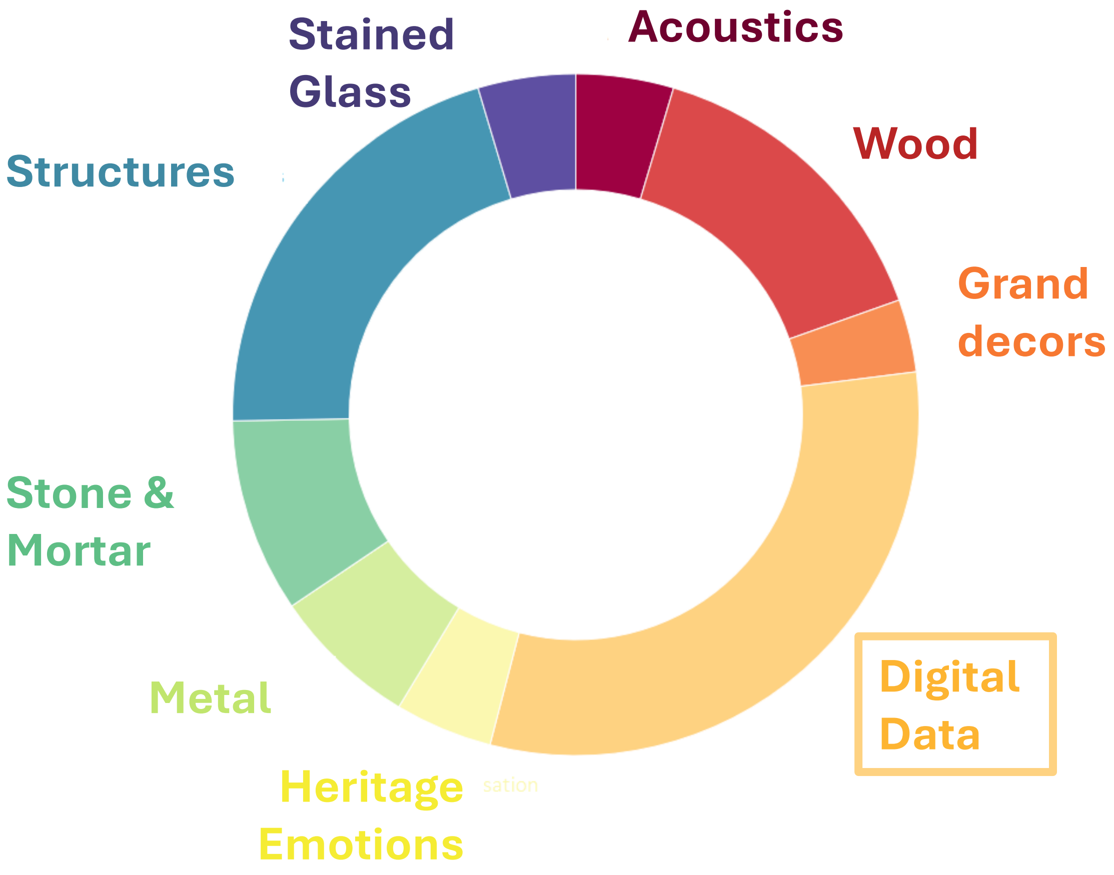


The thematic Working Groups

Scientific Context

- Notre-Dame de Paris, the Scientific mobilisation
  - 9 thematic Working Groups (WG)
  - Study, analyse the building and accompany the restoration work

- Digital Data WG
  - Focus on digital data



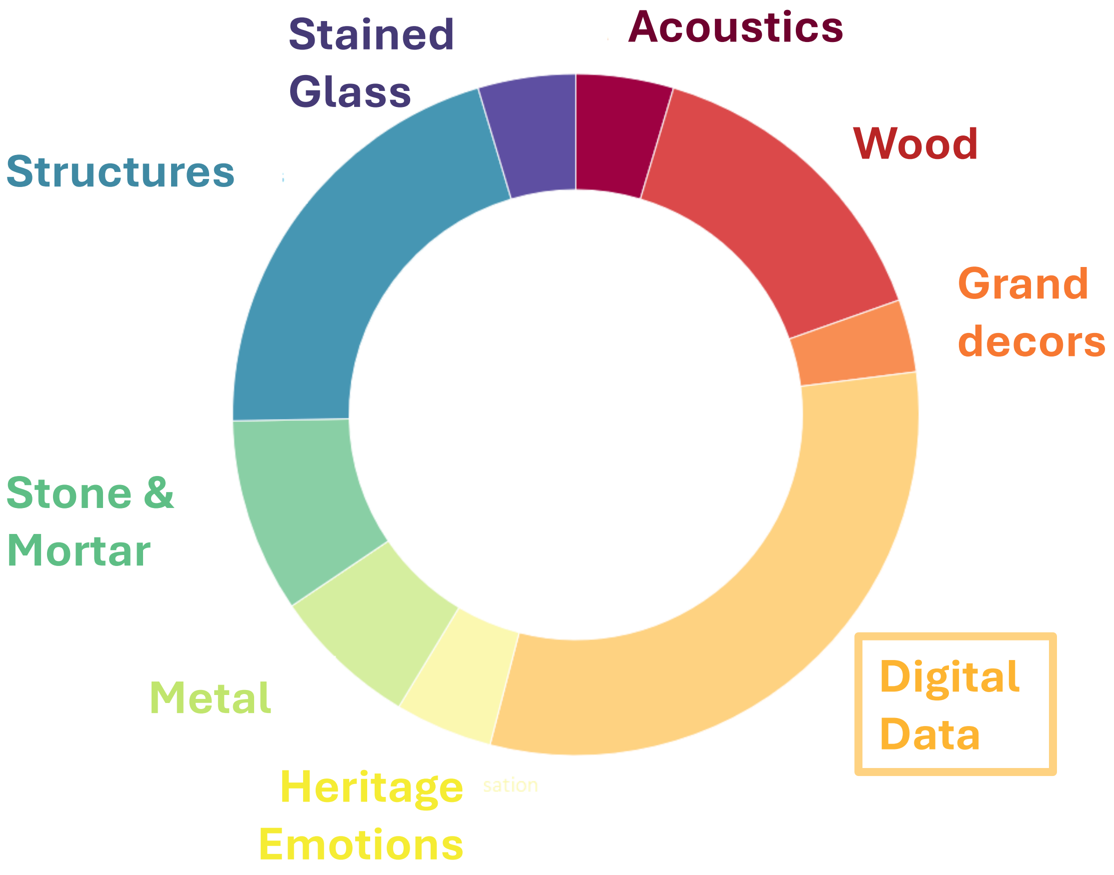
The thematic Working Groups

Scientific Context

- Notre-Dame de Paris, the Scientific mobilisation
  - 9 thematic Working Groups (WG)
  - Study, analyse the building and accompany the restoration work

- Digital Data WG
  - Focus on digital data

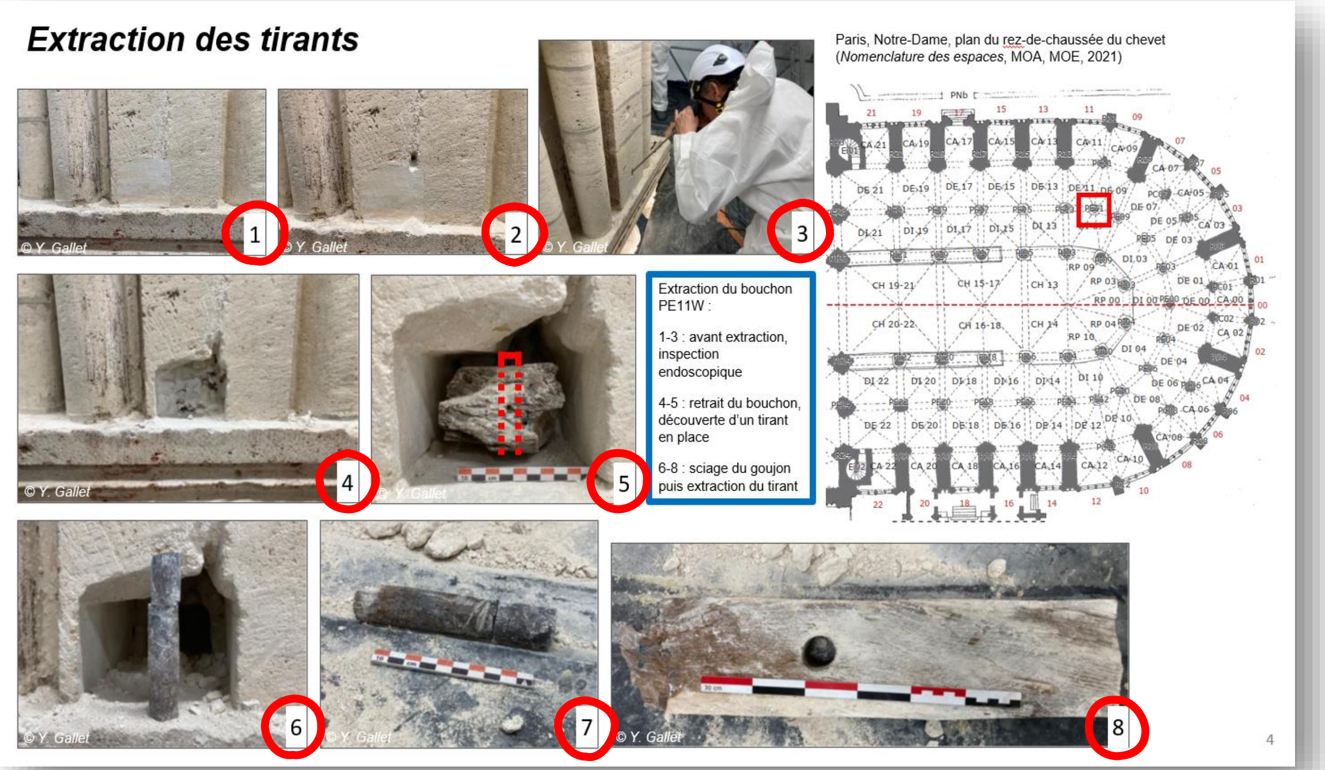
- PhD subject
  - Focus on images



The thematic Working Groups

Problem Statement

- Image uses:
  - Documentation of protocols

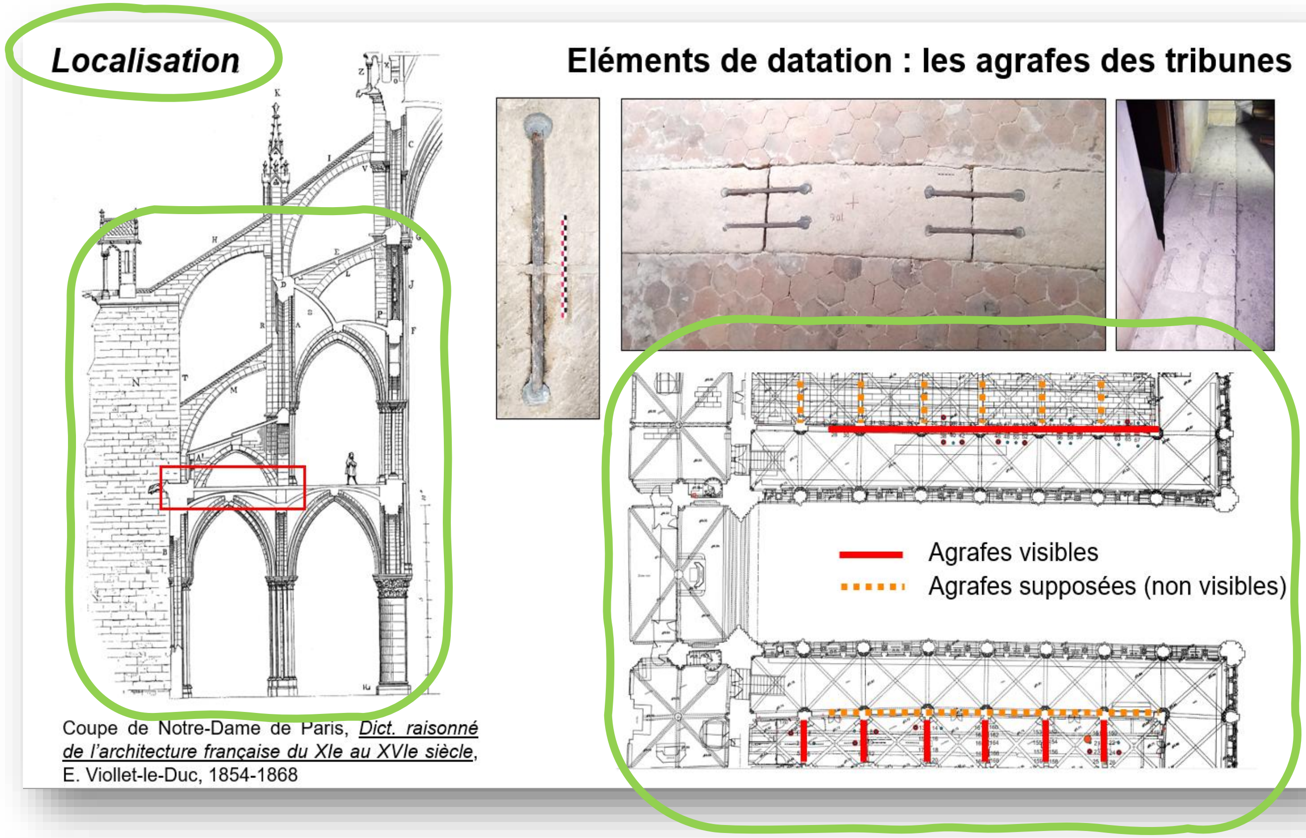


Dater le chevet de Notre-Dame de Paris : regards croisés
Yves Gallet, Jean-Yves Hunot, Mathilde Bernard, 2024

TIME

Problem Statement

- Image uses:
  - Documentation of protocols
  - Location of artefacts

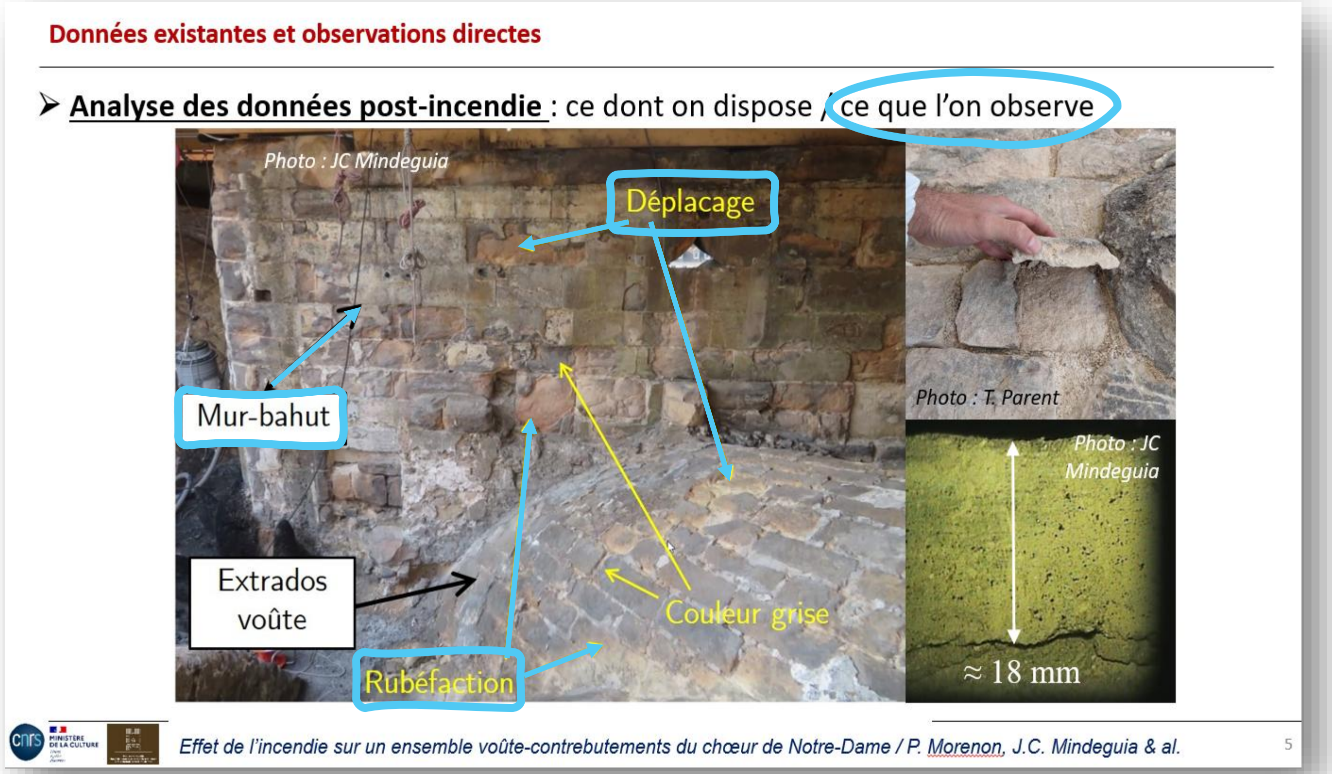


La construction de la nef : nouvelles données chronologiques, Arnaud Ybert, Maxime L'Héritier, Olivier Girardclos , 2024

SPACE

Problem Statement

- Image uses:
  - Documentation of protocols
  - Location of artefacts
  - Annotation of observations



Effet de l'incendie sur un ensemble voûte-contrebutements du chœur de Notre-Dame, Pierre Morenon, Jean-Christophe Mindeguia, 2024

VISUAL CONTENT

Problem Statement

- Focus on images:
  - Rich vectors of information
  - Used by multiple actors
  - Large quantity

- Focus on 3 axes:
  - Time
  - Space
  - Visual Content

Problem Statement

- Focus on images:
  - Rich vectors of information
  - Used by multiple actors
  - **Large quantity**

- Focus on 3 axes:
  - Time
  - Space
  - Visual Content

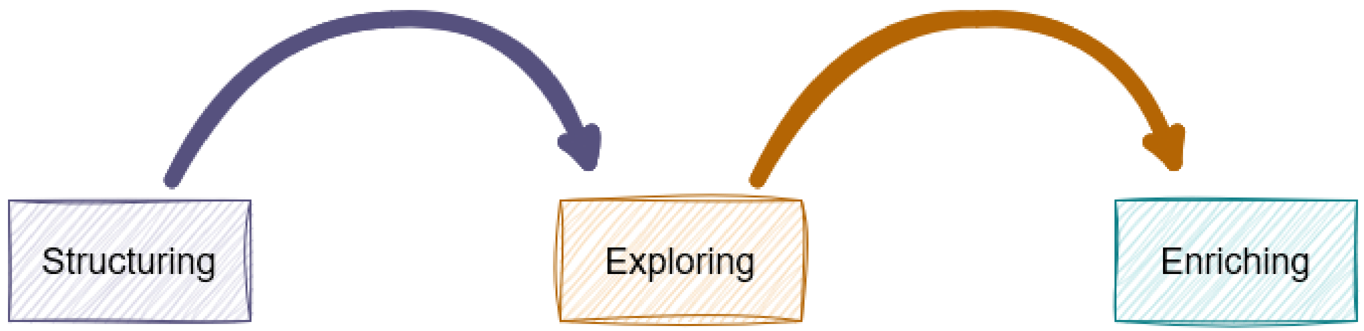
119,864

Research Questions

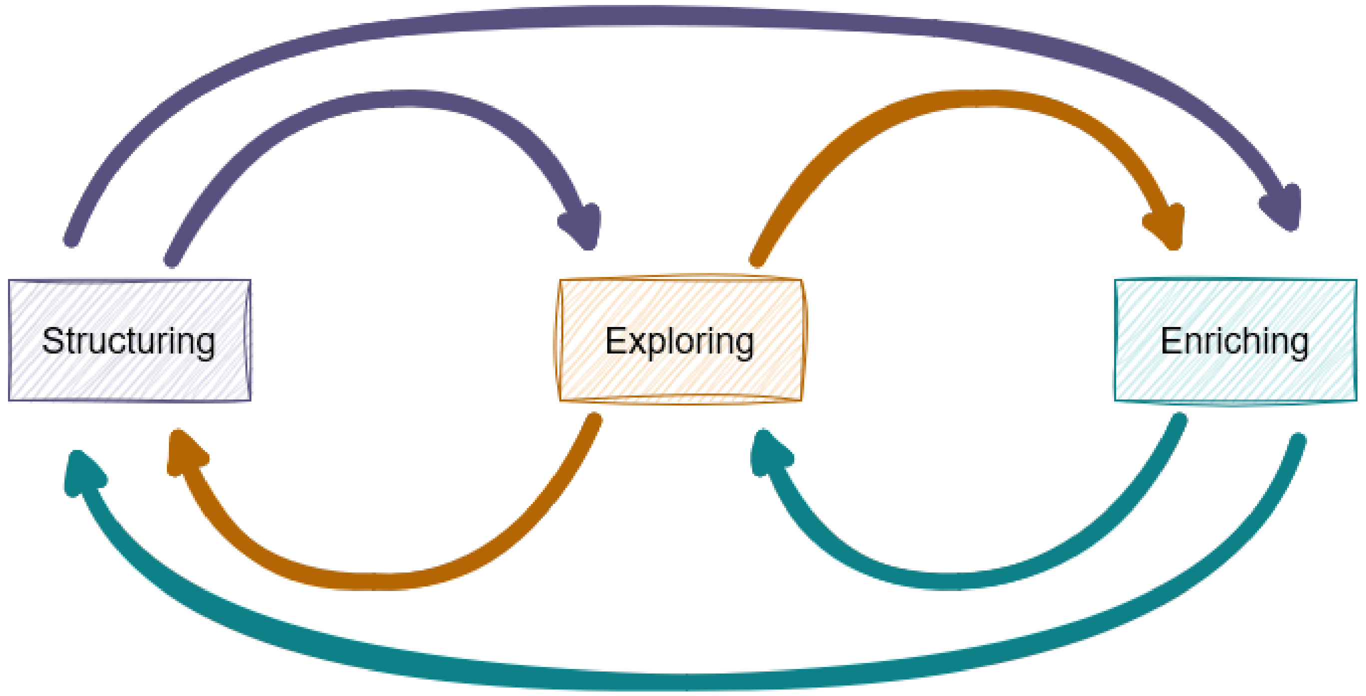
- How to manage this corpus of images, with regard to their temporal, spatial and visual characteristics?

- How to navigate this corpus of images?
  - How to measure (temporal, spatial, visual) ***distances*** between images?

Process



Process



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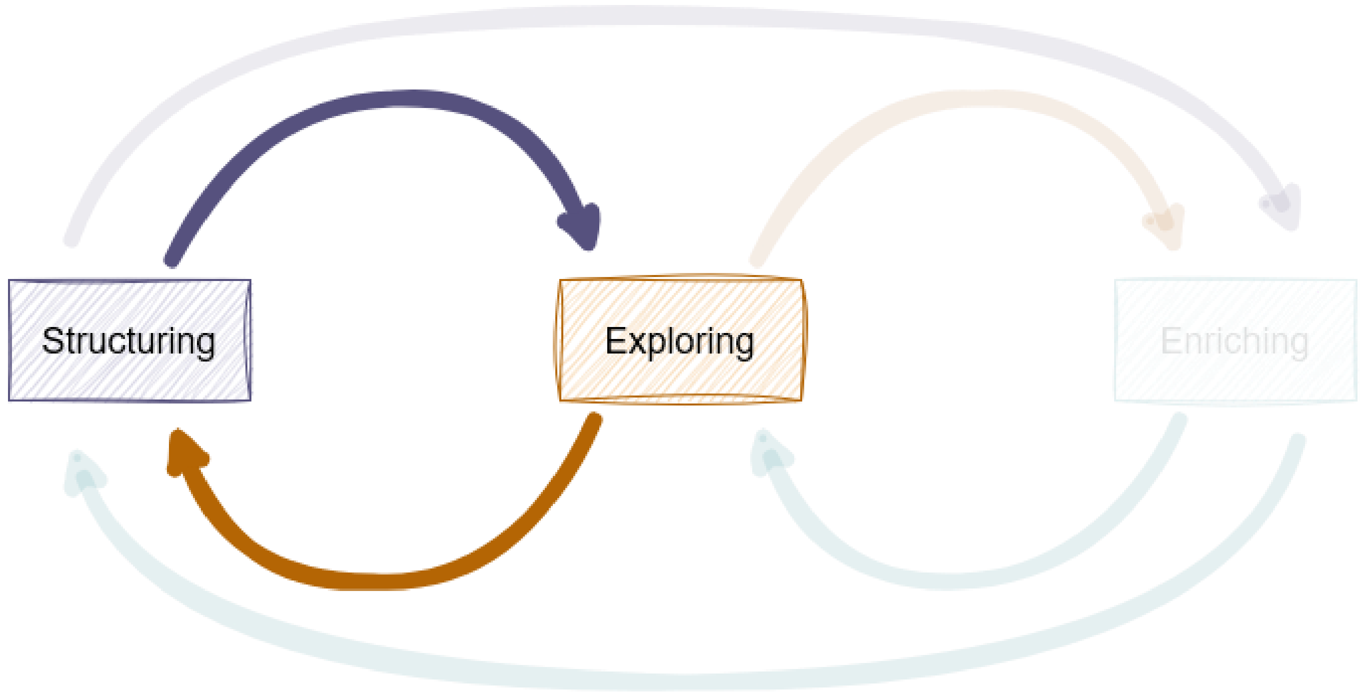
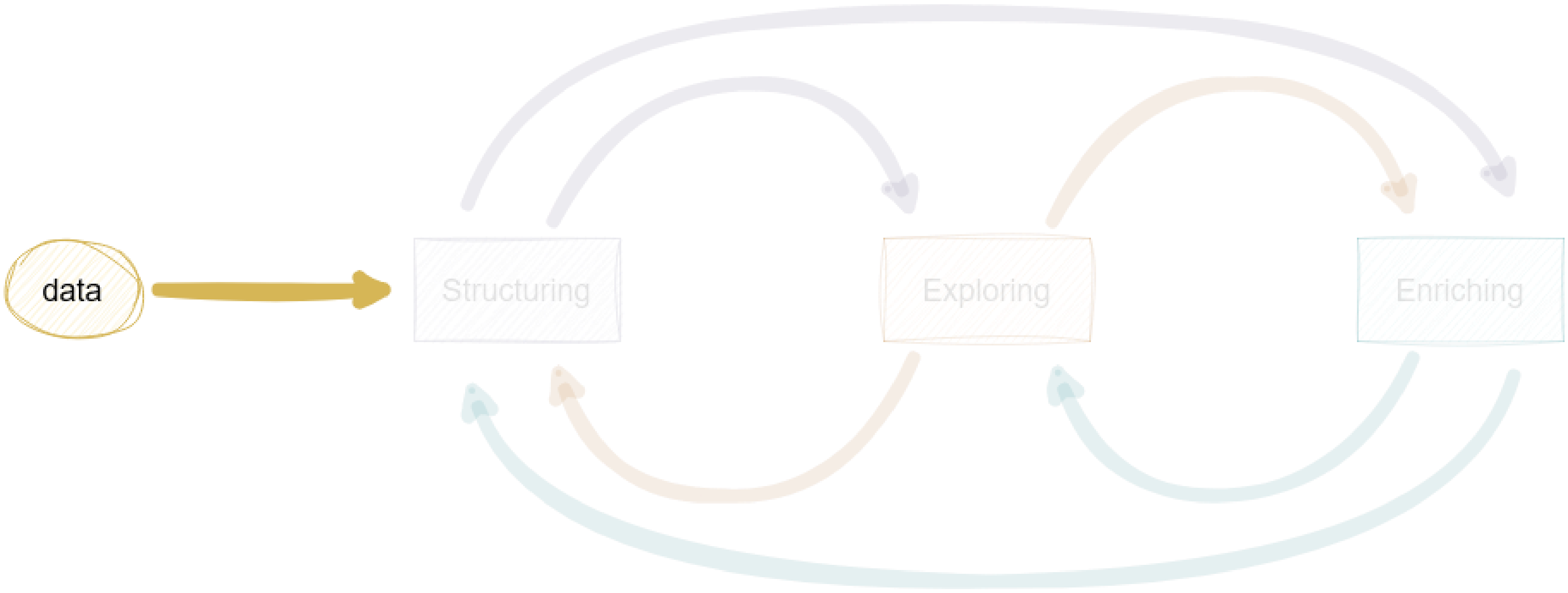
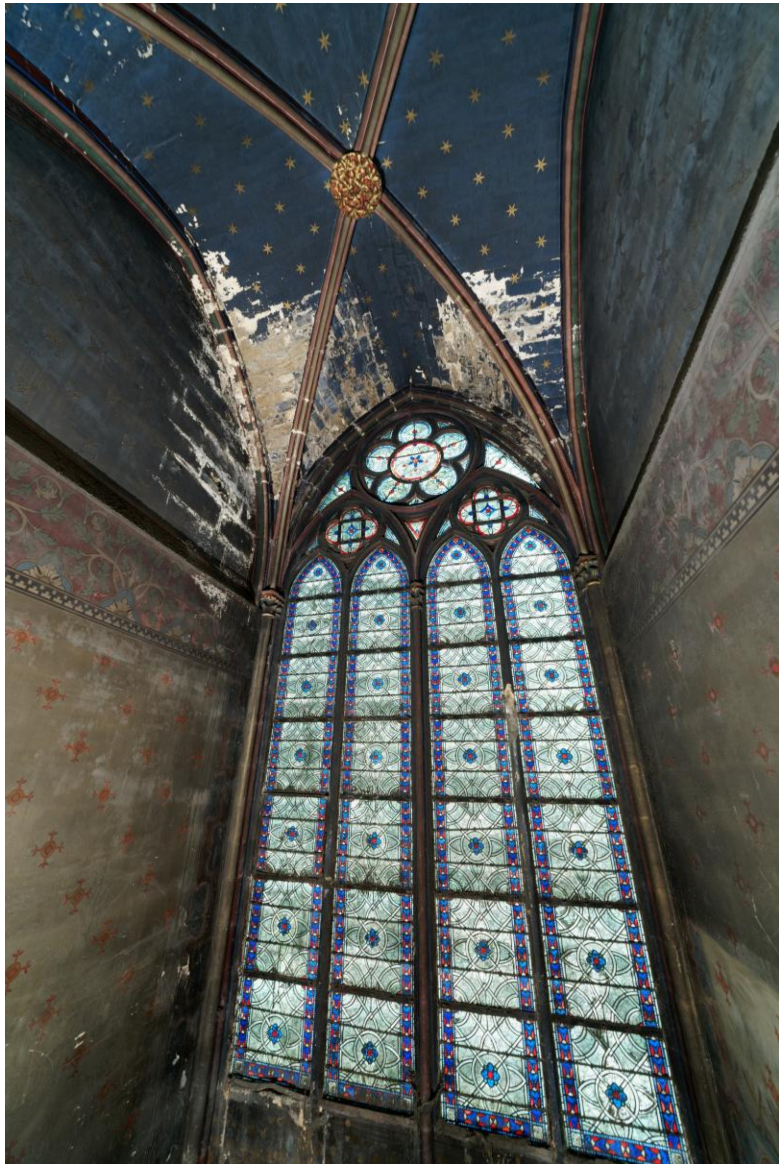


Process



Data

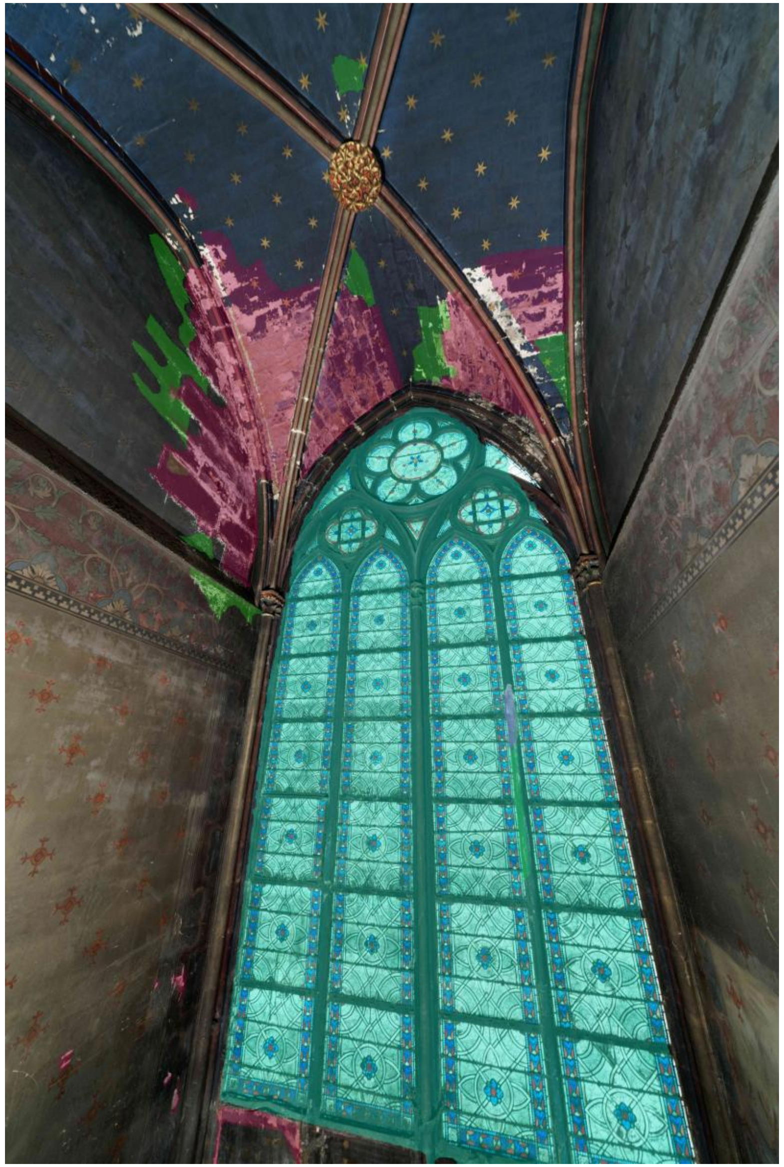
- Visual content



ITER_ca_19_04642

Data

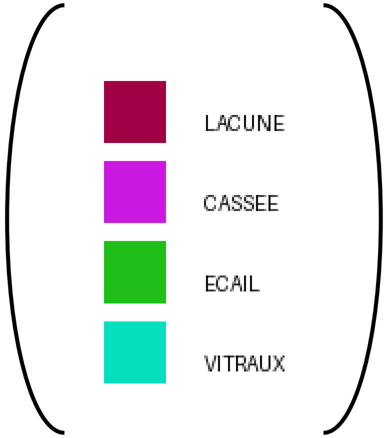
- Visual content

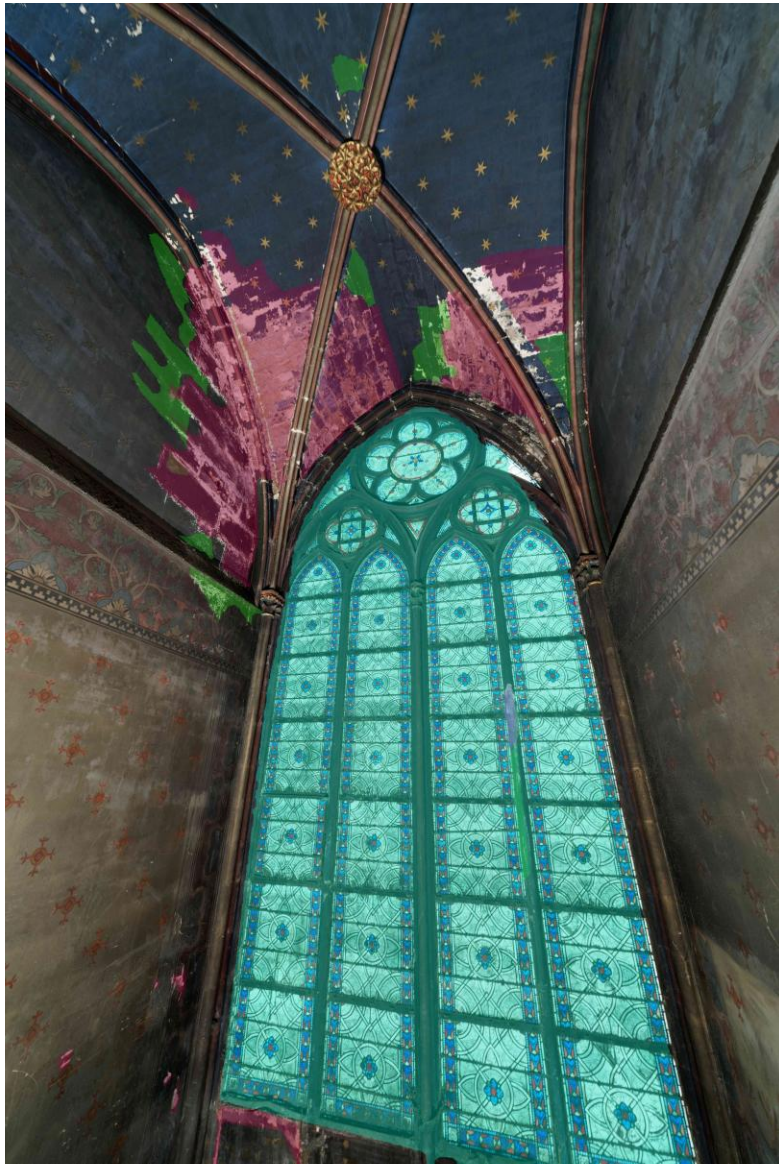


ITER_ca_19_04642

Data

- Visual content

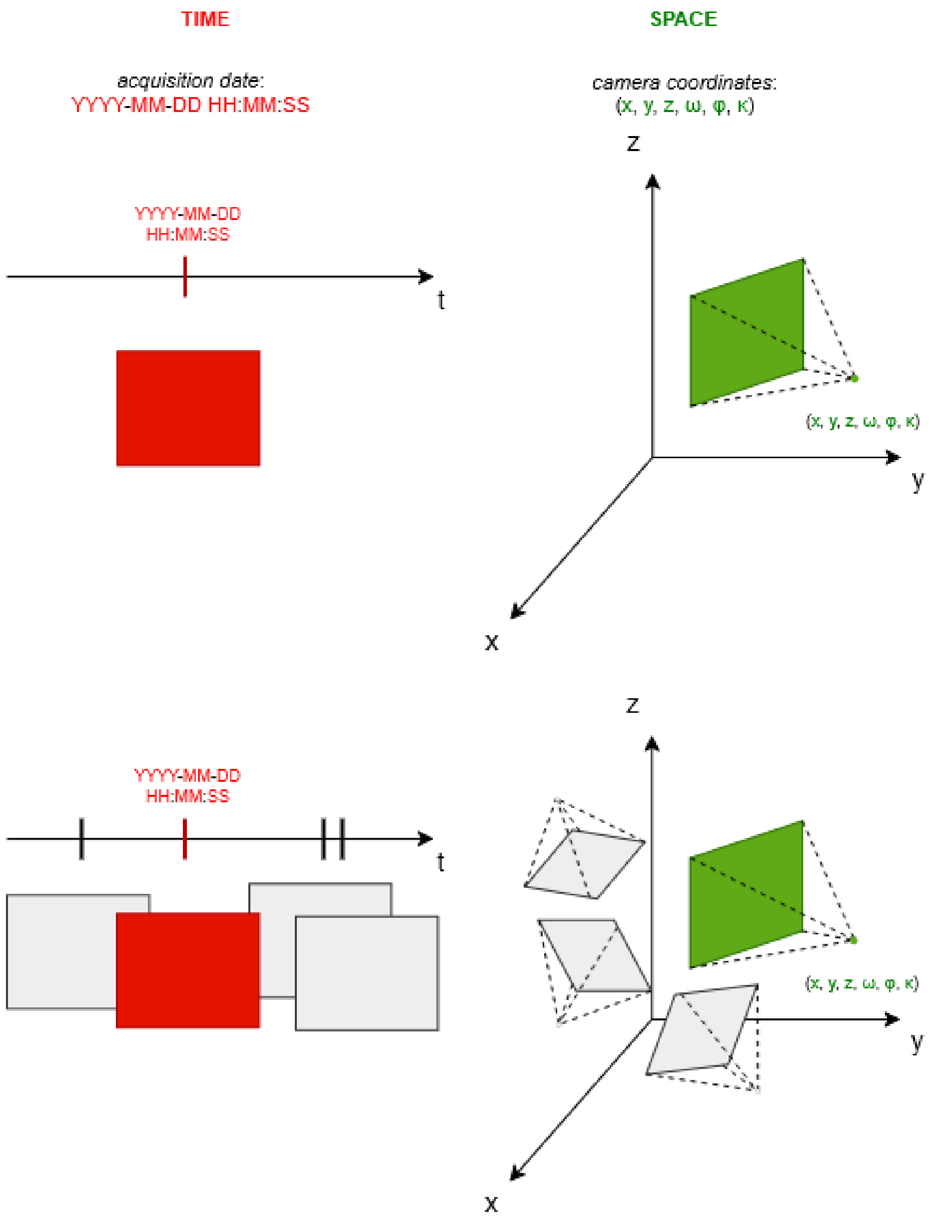




ITER_ca_19_04642

Data

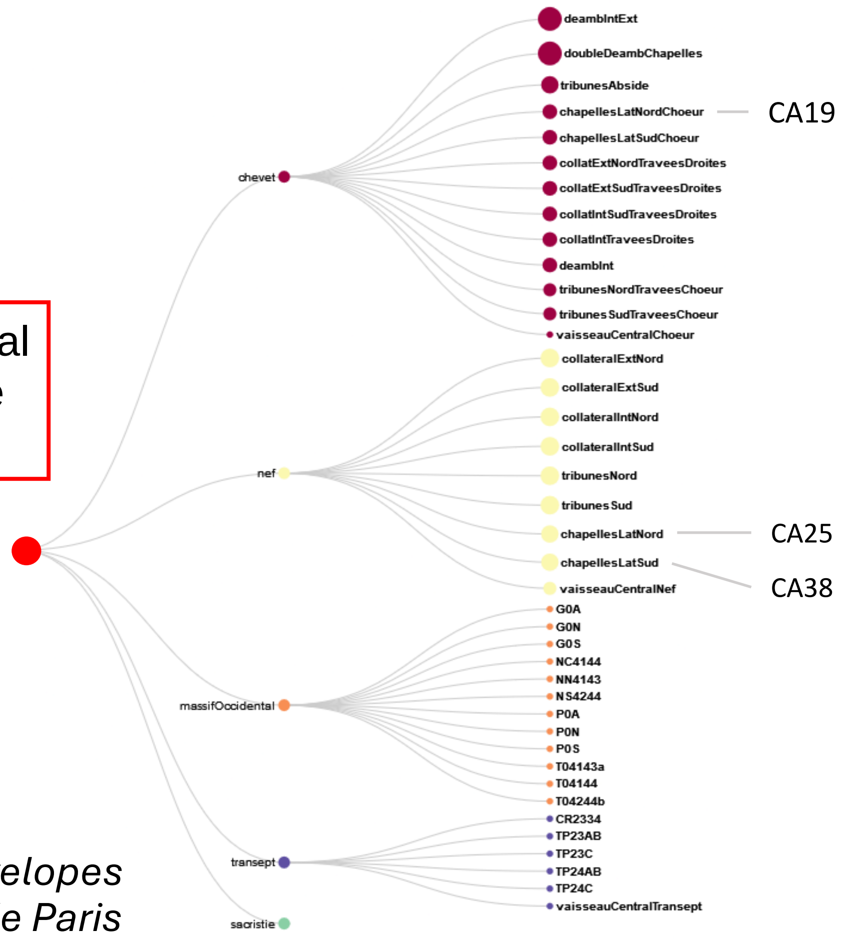
- Visual content
- Image-related metadata



Data

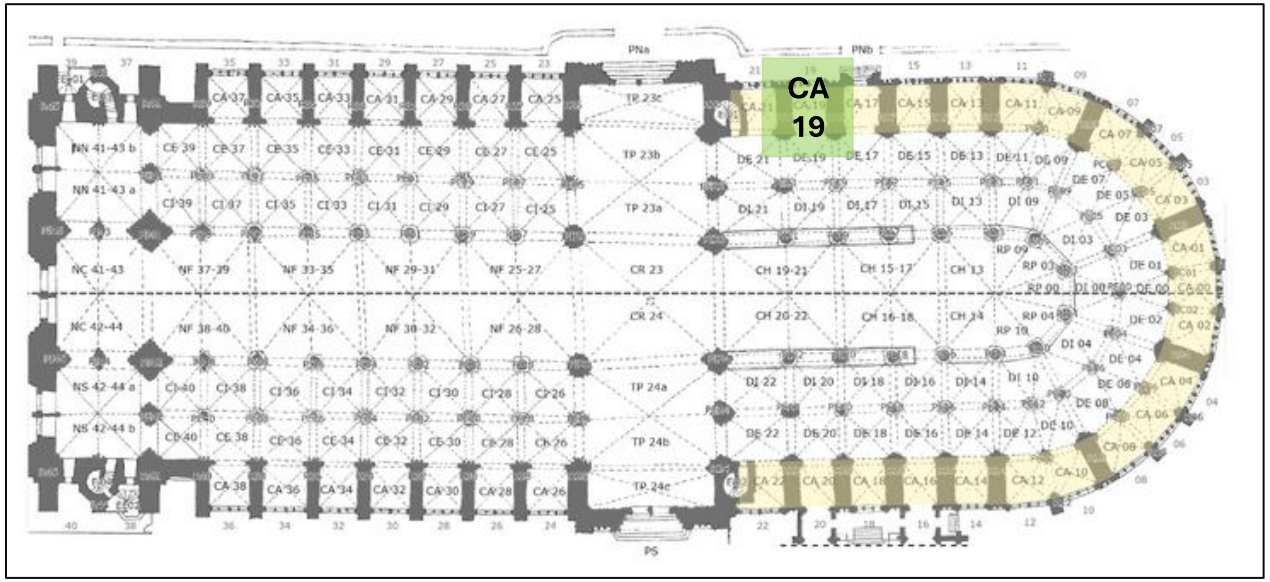
- Visual content
- Image-related metadata
- Corpus-related metadata

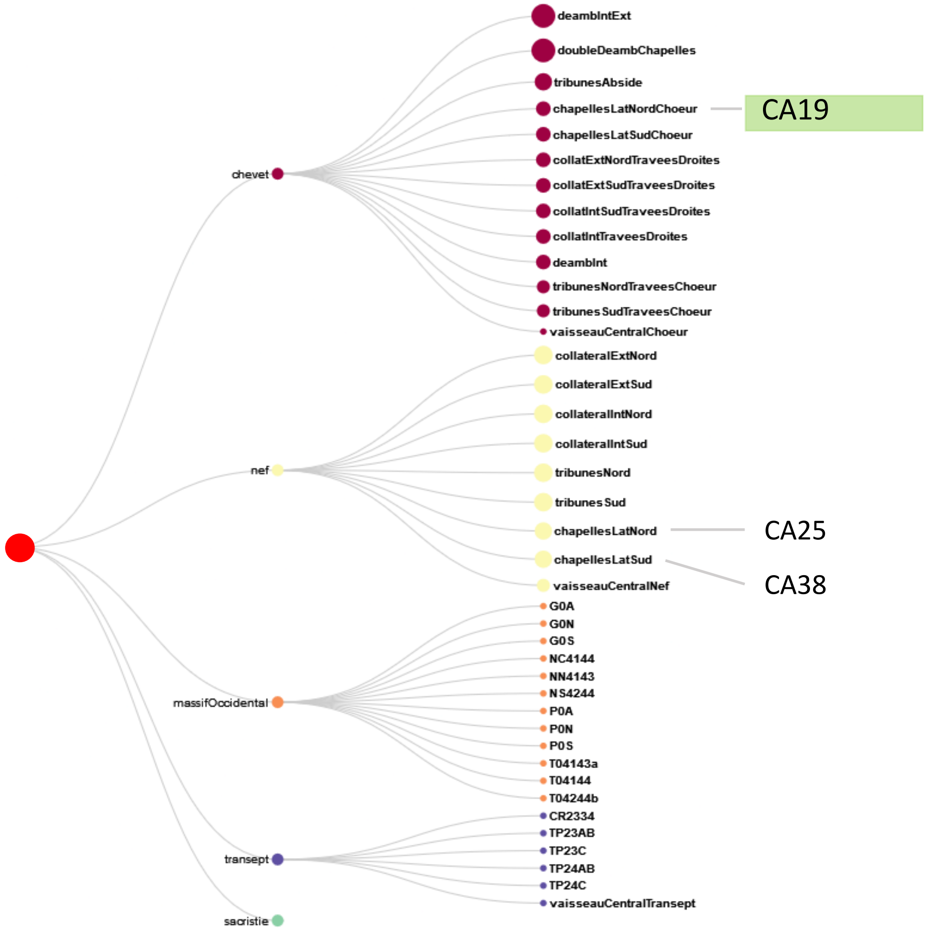
root: cathedral Notre-Dame de Paris



Taxonomy of the spatial envelopes describing the spaces of Notre-Dame de Paris

Data


CA
19


chevet
deambIntExt
doubleDeambChapelles
tribunesAbside
chapellesLatNordChoeur
CA19
chapellesLatSudChoeur
collatExtNordTraveesDroites
collatExtSudTraveesDroites
collatIntSudTraveesDroites
collatIntTraveesDroites
deambInt
tribunesNordTraveesChoeur
tribunesSudTraveesChoeur
vaisseauCentralChoeur
nef
collateralExtNord
collateralExtSud
collateralIntNord
collateralIntSud
tribunesNord
tribunesSud
chapellesLatNord
CA25
chapellesLatSud
CA38
vaisseauCentralNef
massifOccidental
G0A
G0N
G0S
NC4144
NN4143
NS4244
P0A
P0N
P0S
T04143a
T04144
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transept
CR2334
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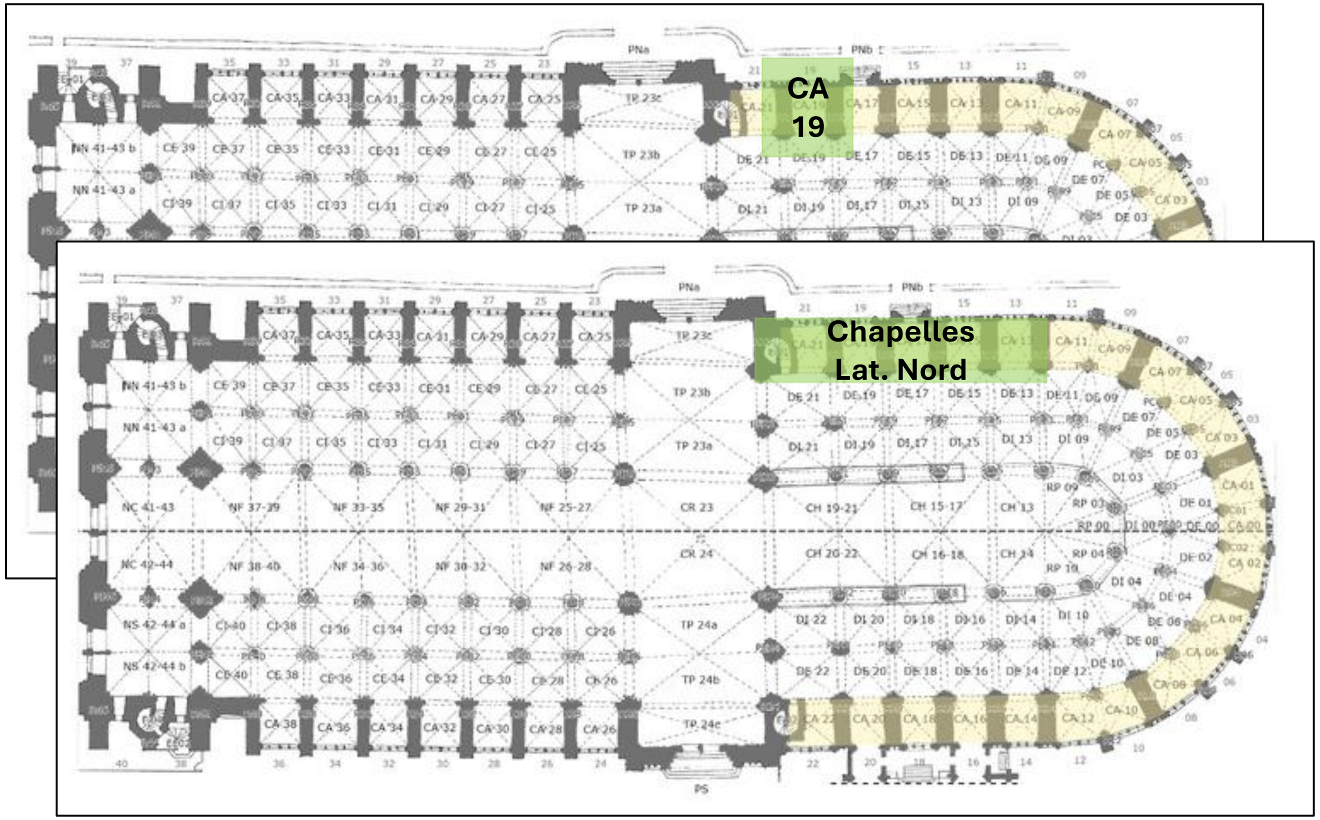
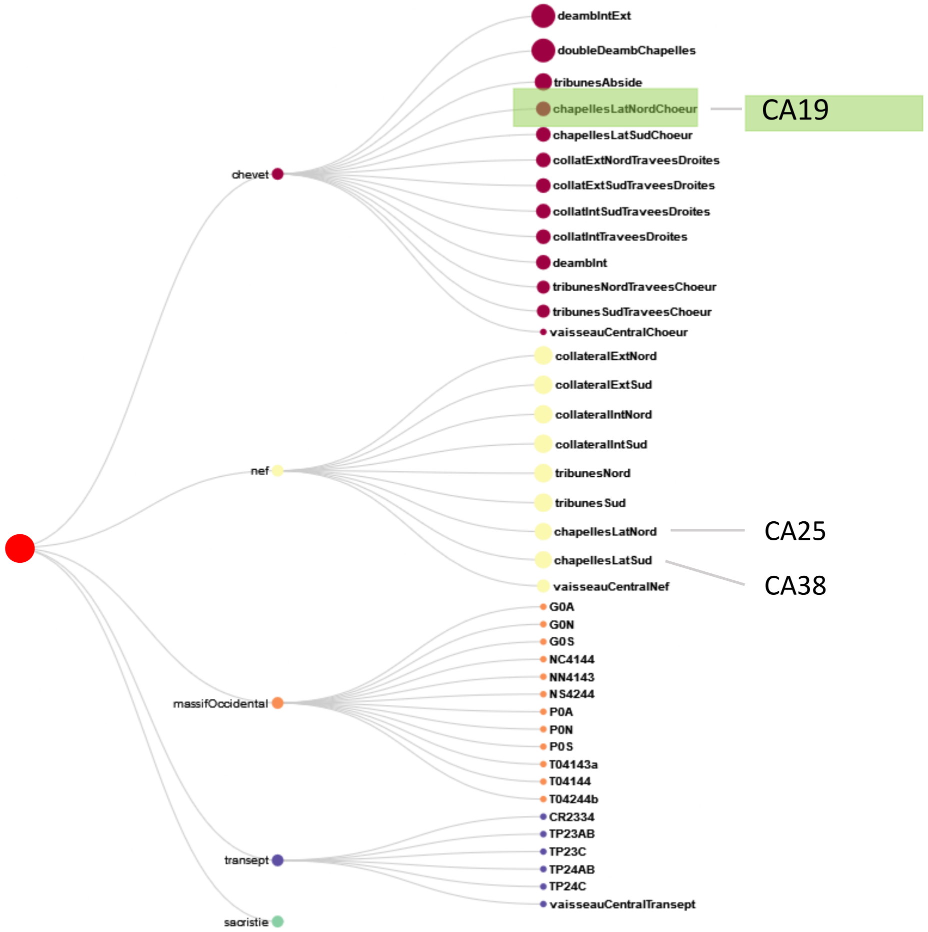
TP23C
TP24AB
TP24C
vaisseauCentralTransept
sacristie

Data


CA
19
Chapelles
Lat. Nord


deambIntExt
doubleDeambChapelles
tribunesAbside
chapellesLatNordChoeur
CA19
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collatExtSudTraveesDroites
collatIntSudTraveesDroites
collatIntTraveesDroites
deambInt
tribunesNordTraveesChoeur
tribunes SudTraveesChoeur
vaisseauCentralChoeur
chevet
collateralExtNord
collateralExtSud
collateralIntNord
collateralIntSud
tribunesNord
tribunes Sud
chapellesLatNord
CA25
chapellesLatSud
CA38
vaisseauCentralNef
nef
G0A
G0N
G0S
NC4144
NN4143
NS4244
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P0N
P0S
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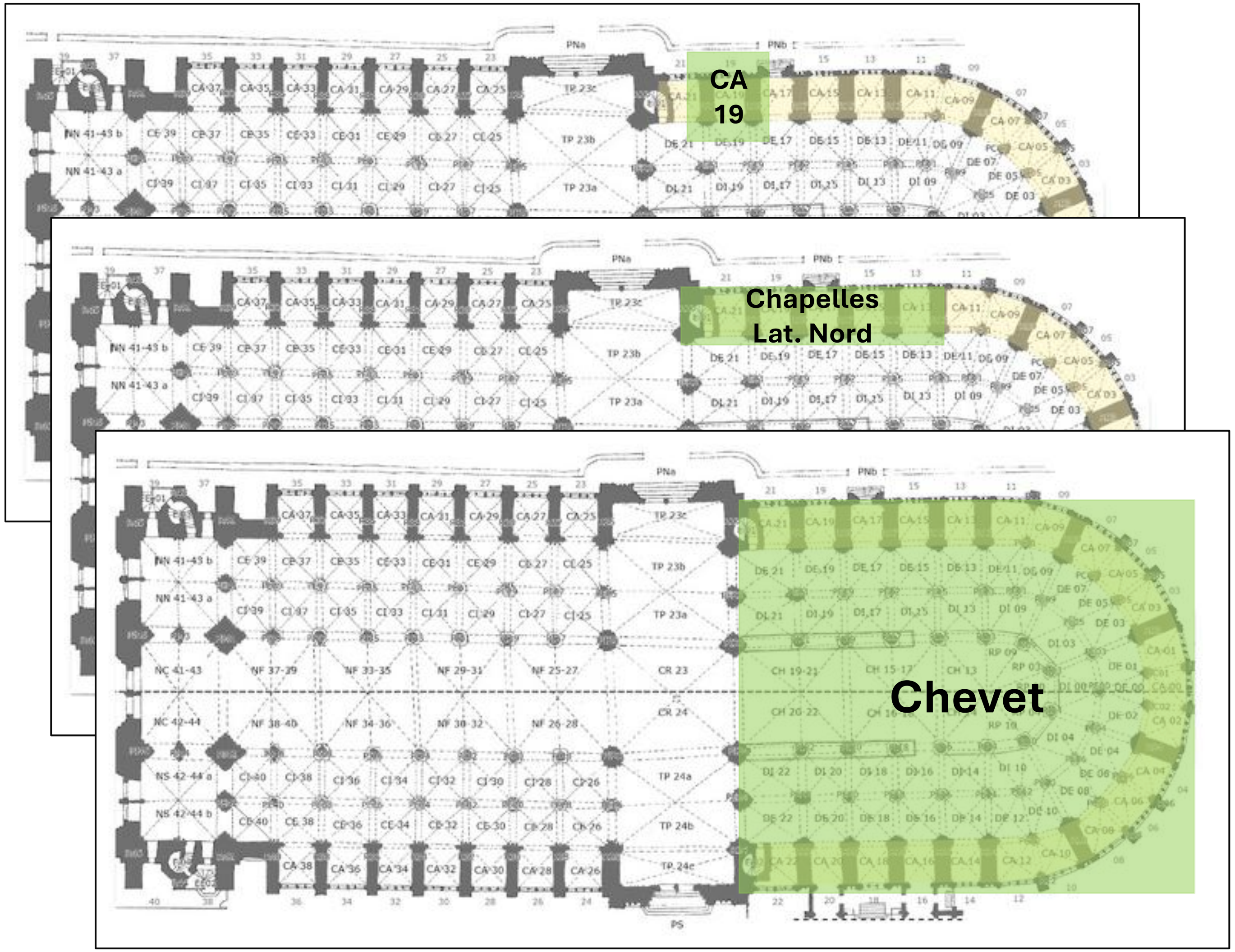
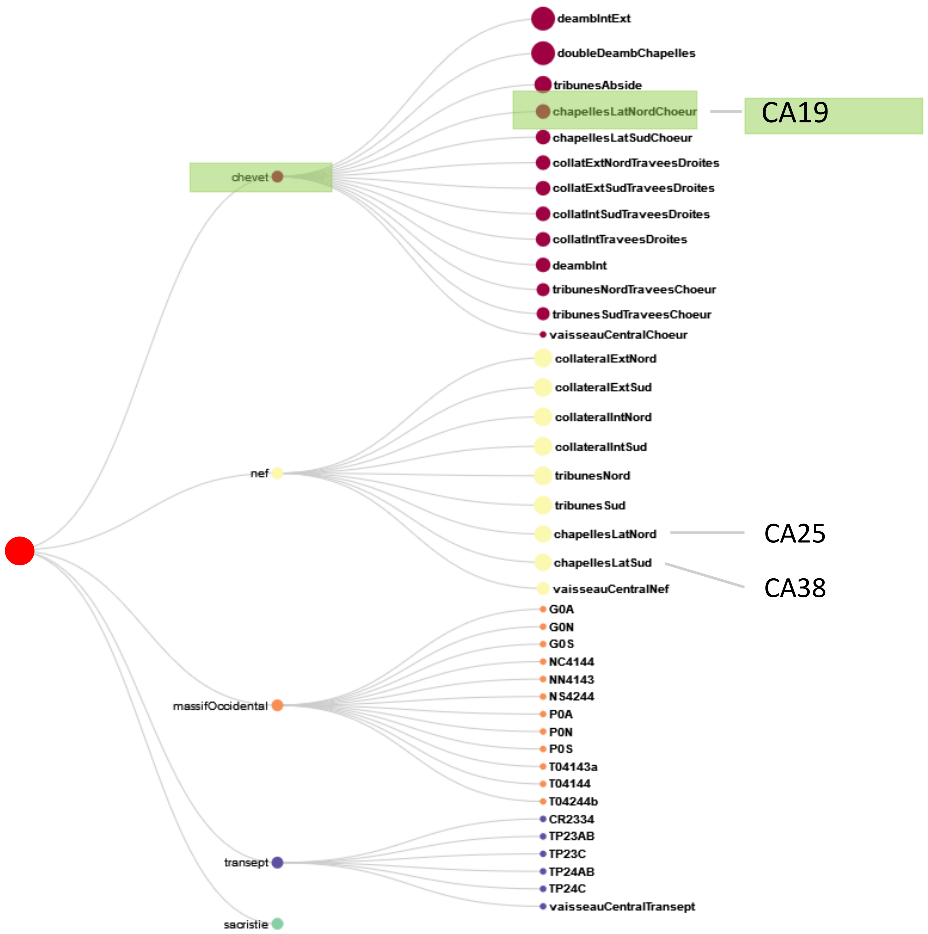
Data


CA
19
Chapelles
Lat. Nord
Chevet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doubleDeambChapelles
tribunesAbside
chapellesLatNordChoeur
CA19
chapellesLatSudChoeur
collatExtNordTraveesDroites
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collatIntTraveesDroites
deambInt
tribunesNordTraveesChoeur
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CA25
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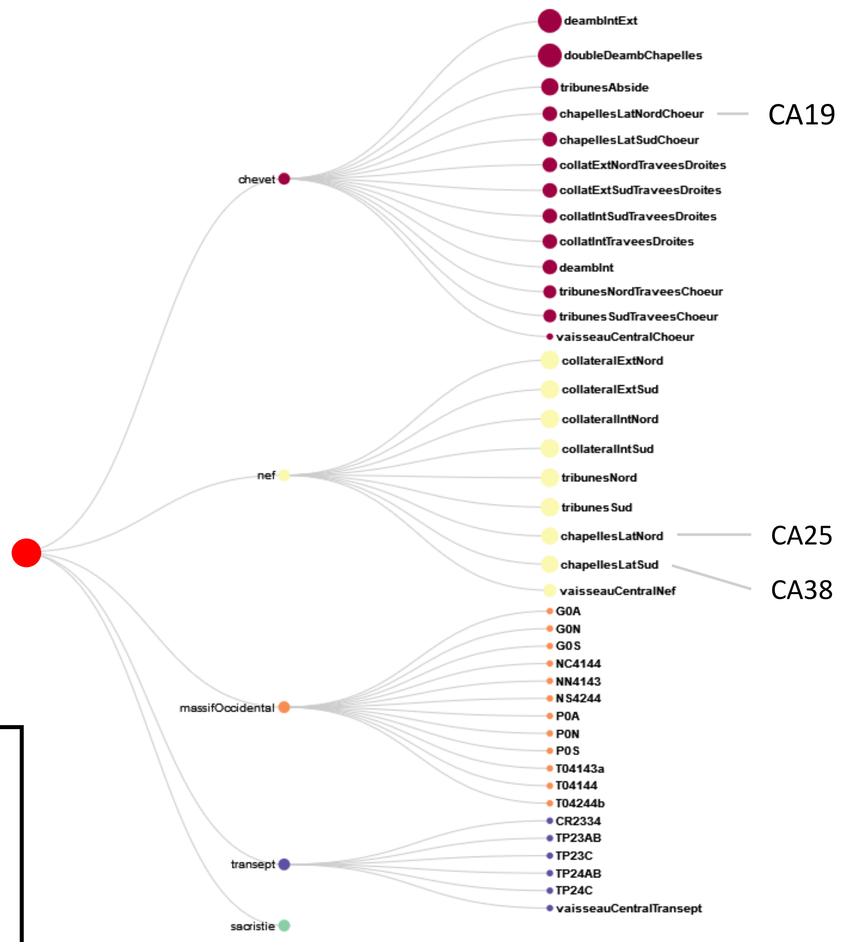
T04244b
massifOccidental
CR2334
TP23AB
TP23C
TP24AB
TP24C
vaisseauCentralTransept
transept
sacristie

Data

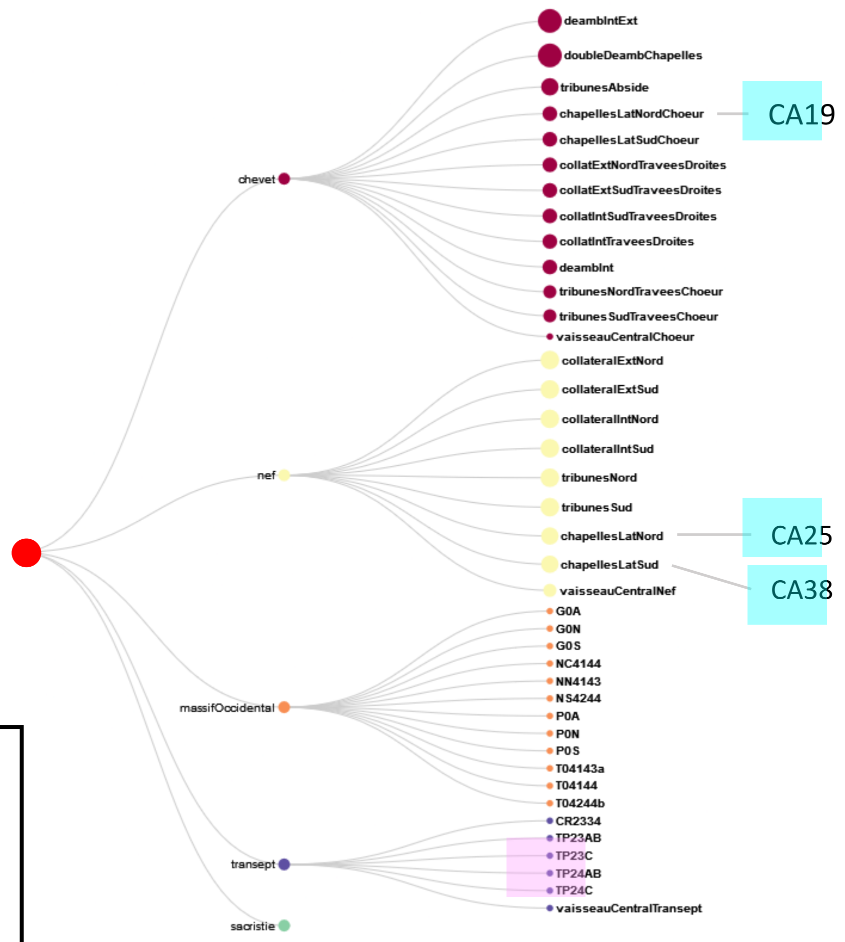
- Visual content
- Image-related metadata
- Corpus-related metadata
- **Generic metadata**



CA: **Chapel**
TP: **Transept**
Sacristy
Nave
…

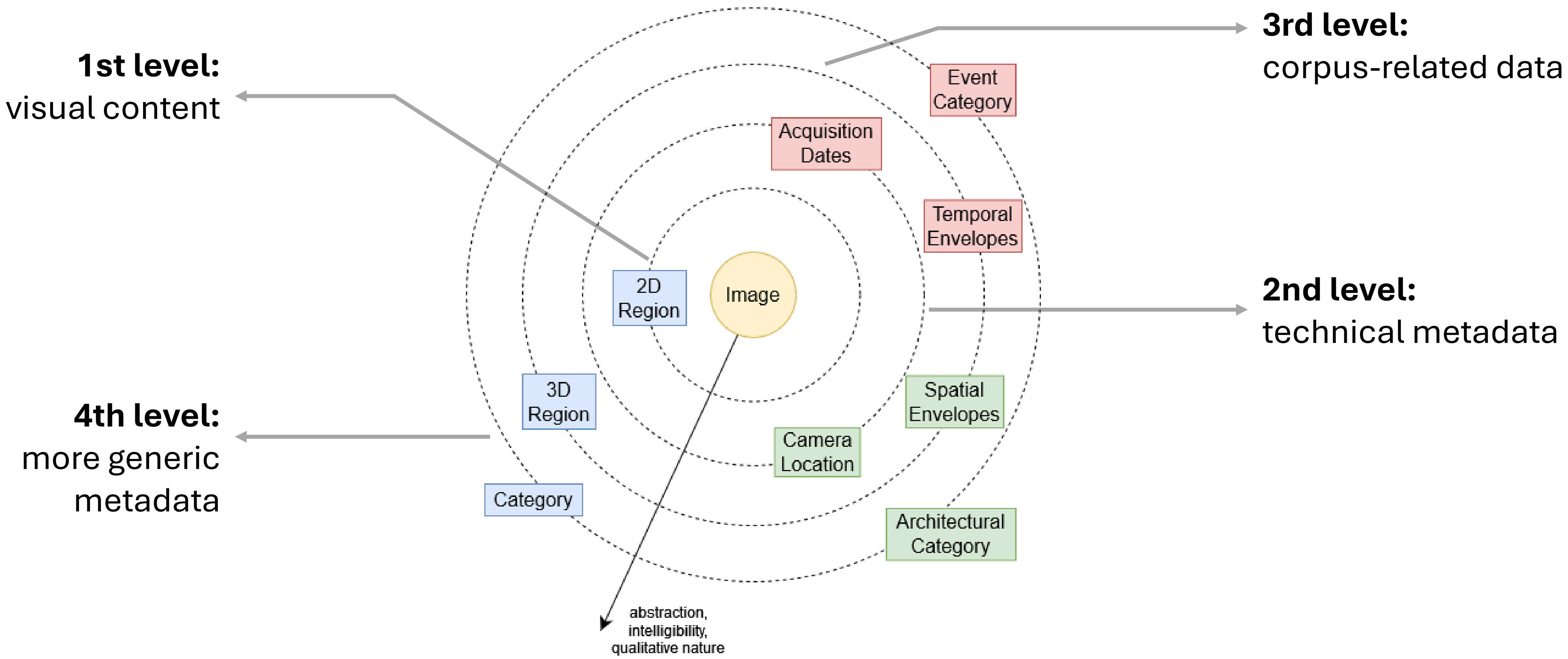
Data

- Visual content
- Image-related metadata
- Corpus-related metadata
- **Generic metadata**



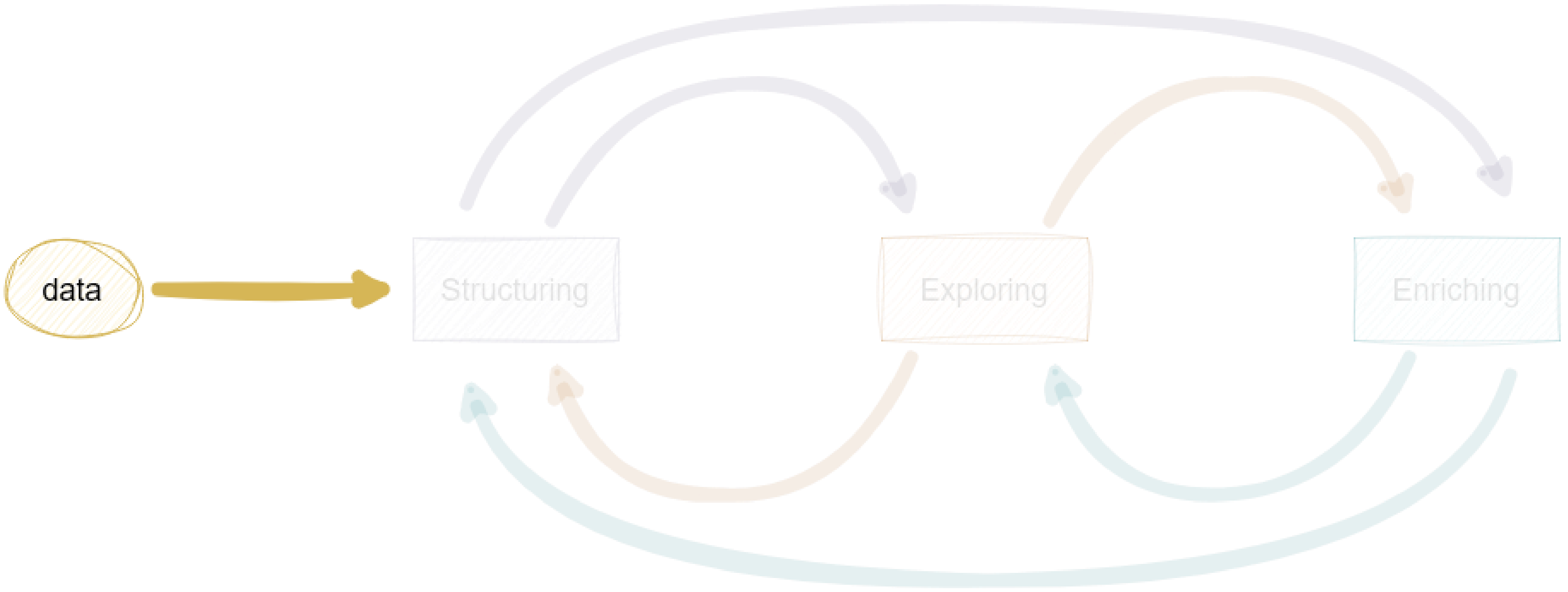
CA: **Chapel**
TP: **Transept**
Sacristy
Nave
…

Data – Summary

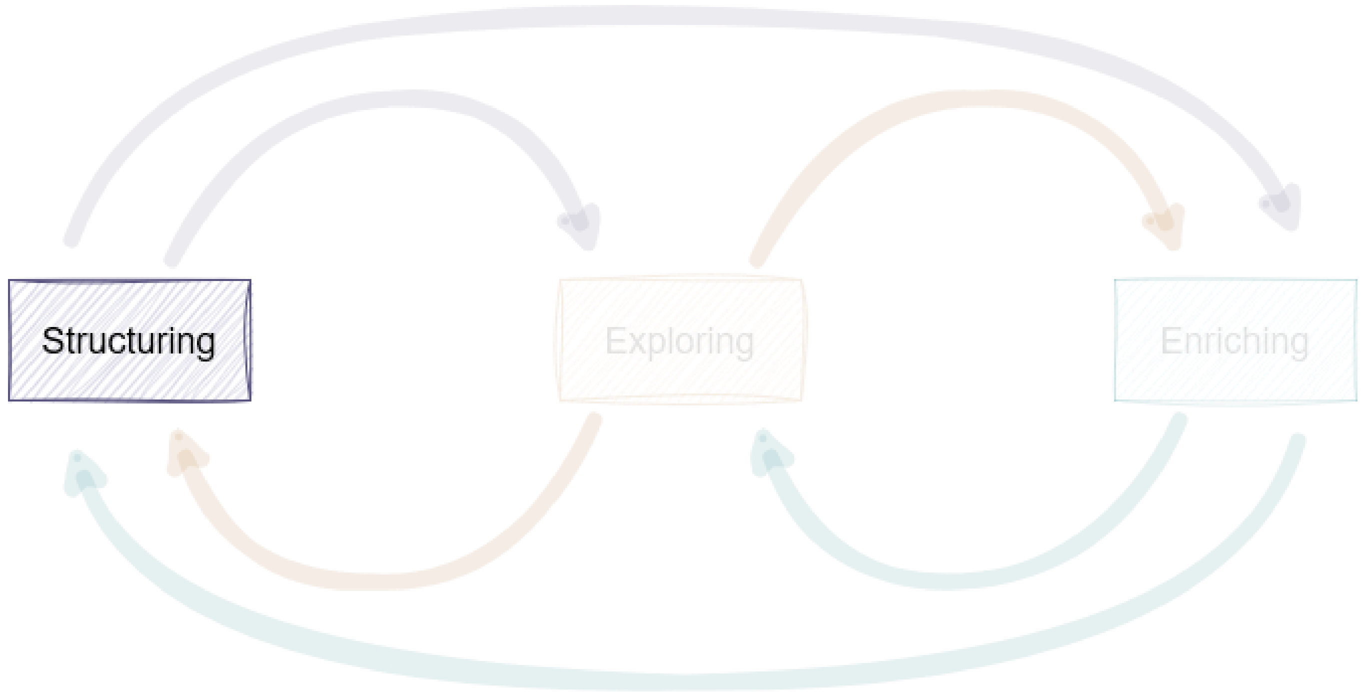


Levels of proximity of the different metadata characterising the images

Process



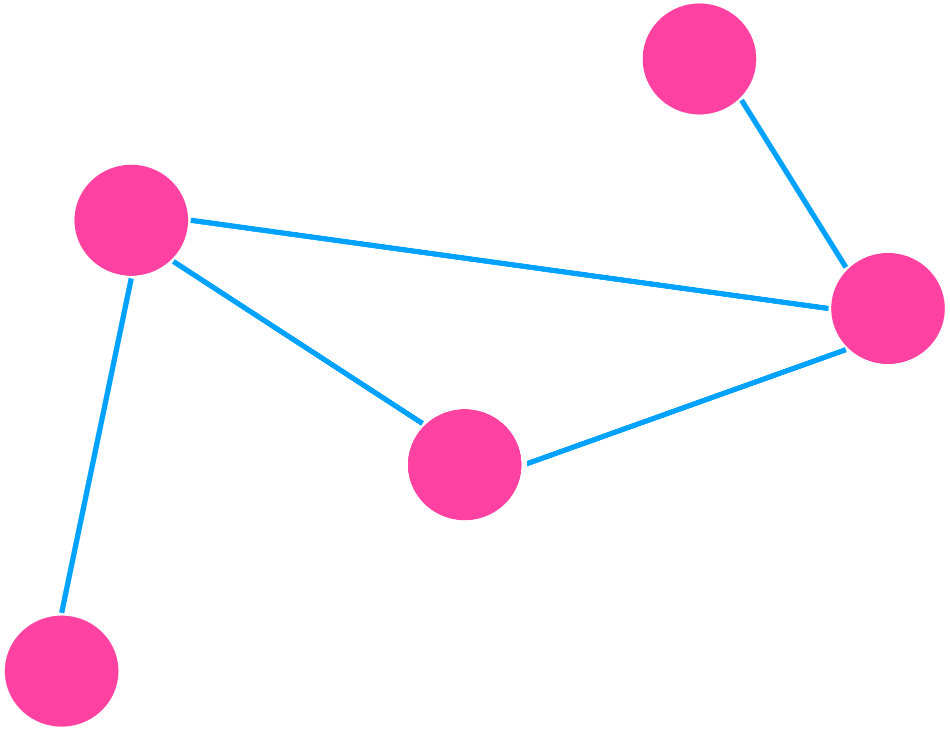
Process



Structuring (I)

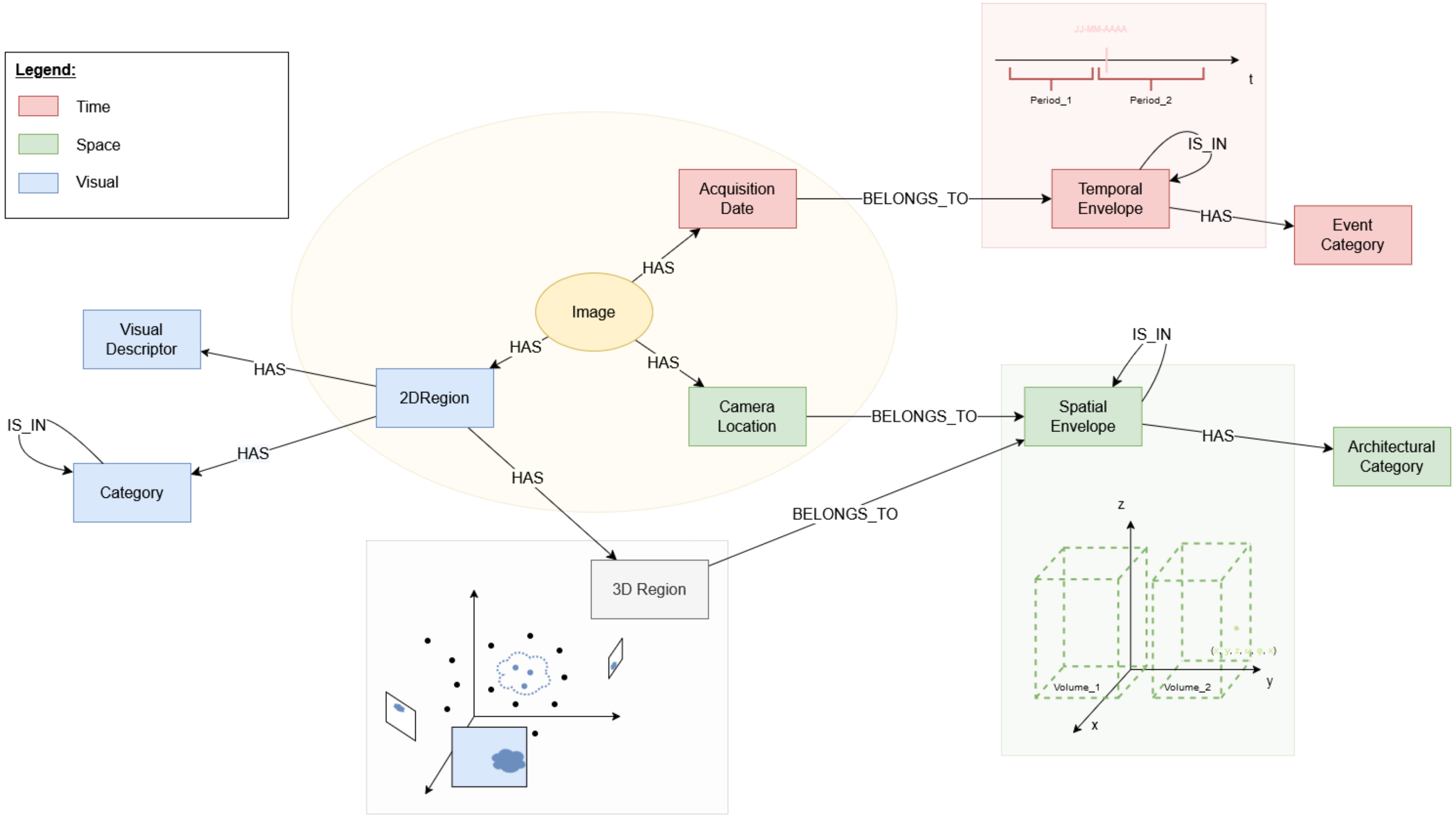
- Graph-oriented database

> *Graphs are a* <u>ubiquitous</u> *data structure [...] a collection of objects (i.e.,* ***nodes****), along with a set of interactions (i.e.,* ***edges****) between pairs of these objects.* **[Hamilton, 2020]**



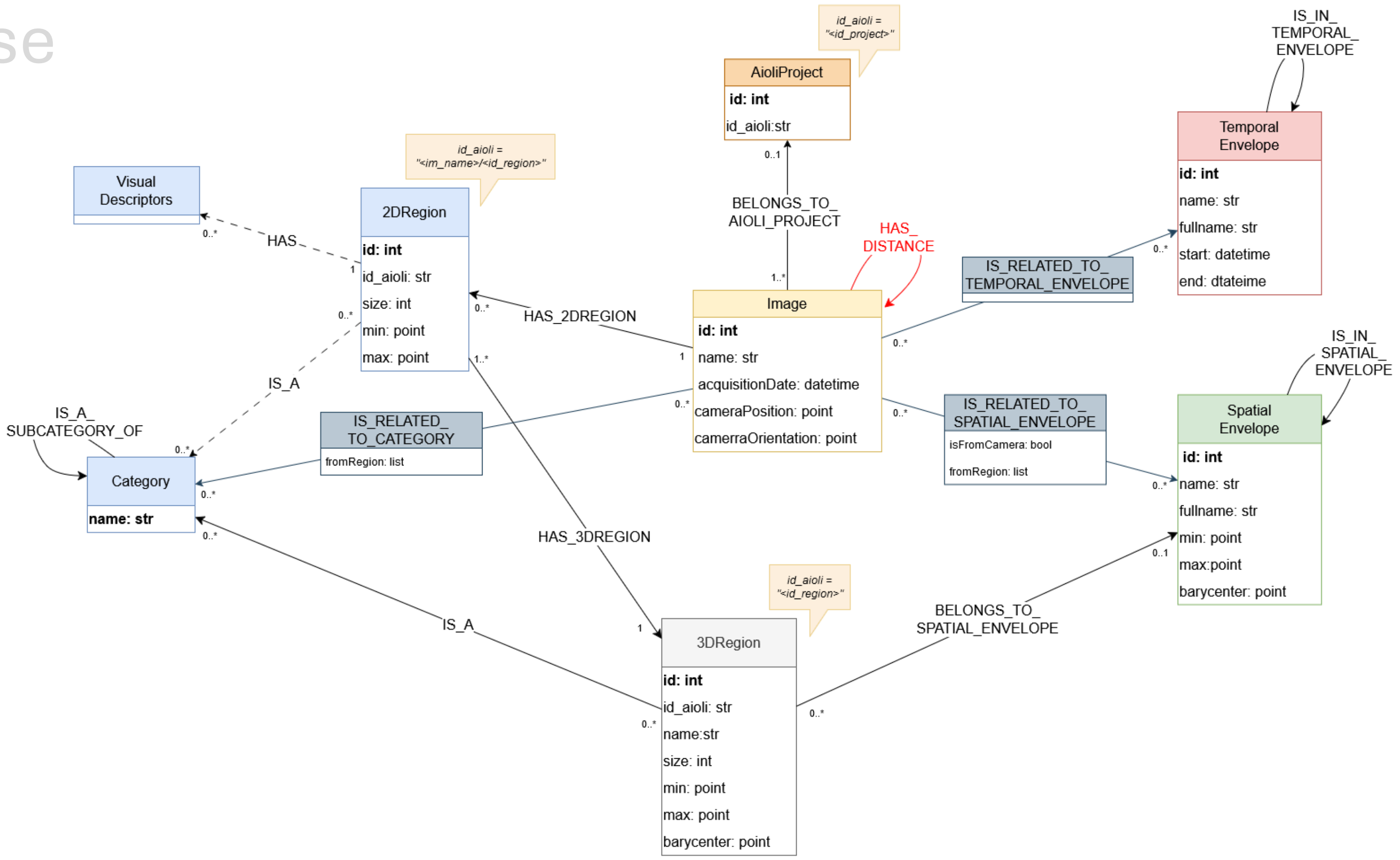
Structuring (I)

- Graph-oriented database
- Data model



Structuring (I)

- Graph-oriented database
- Data model
- Database schema



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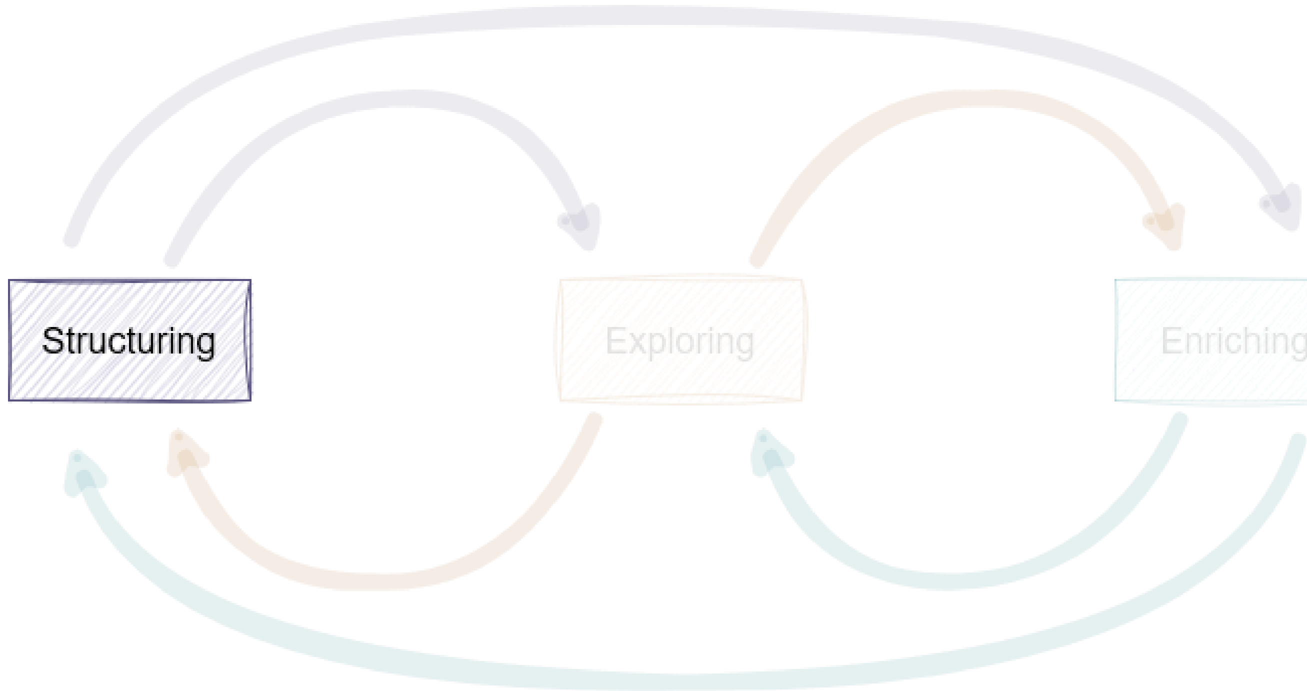
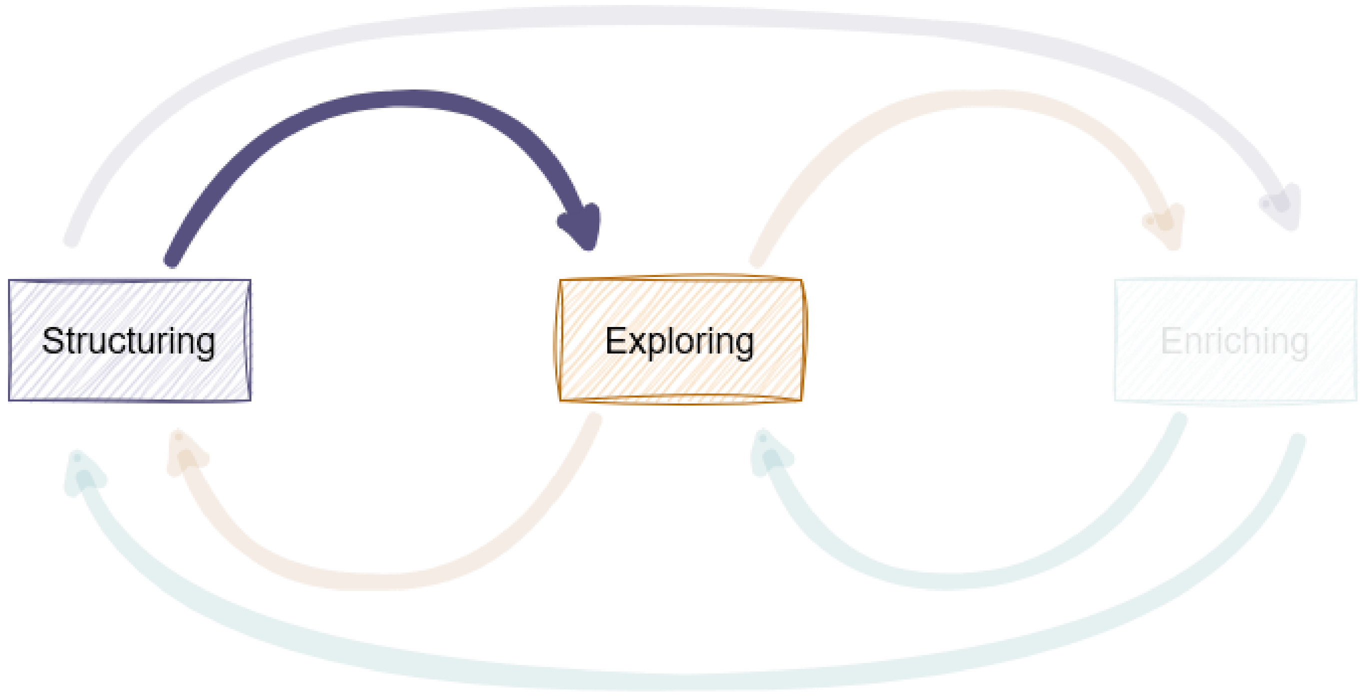
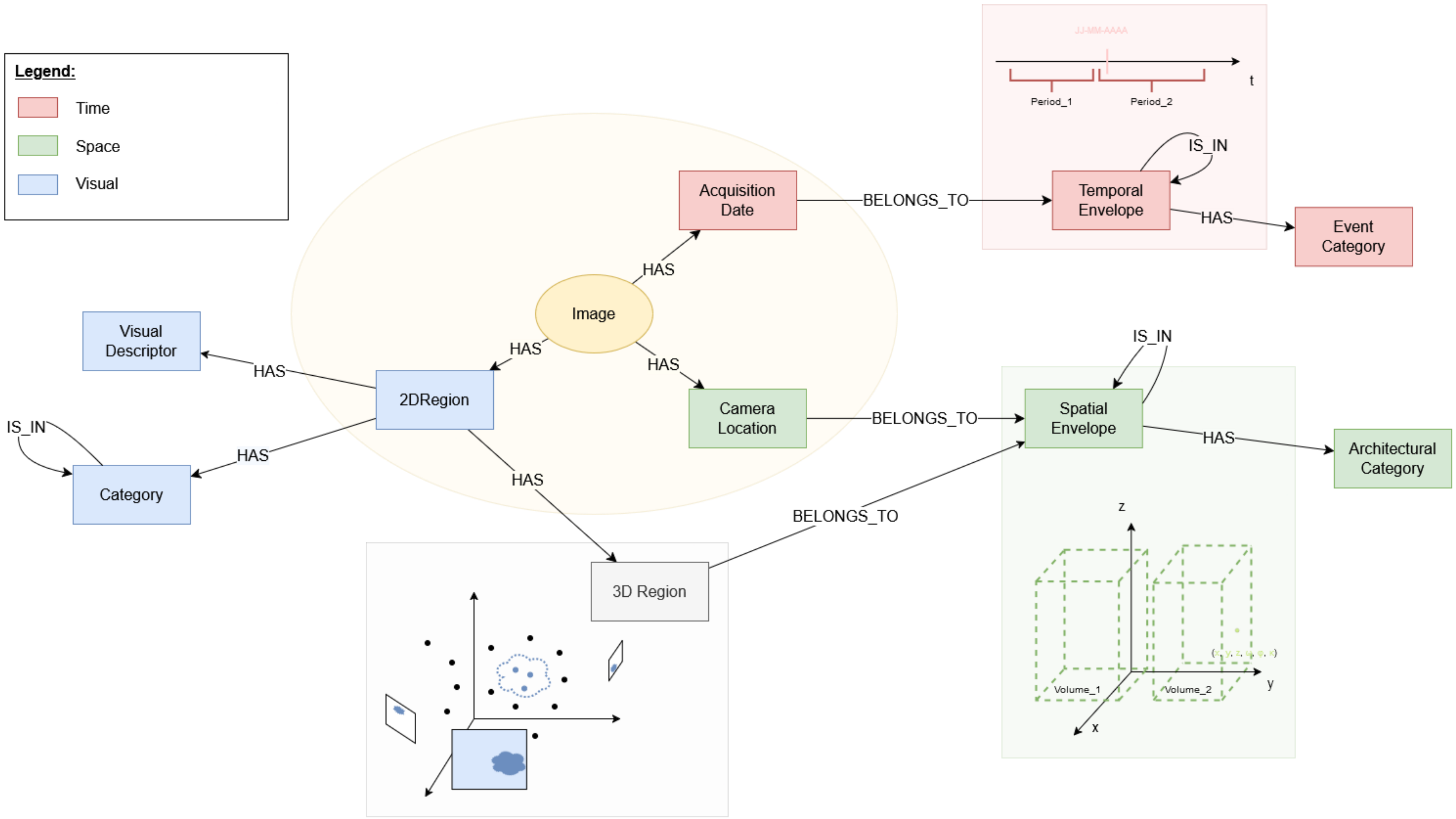


Process

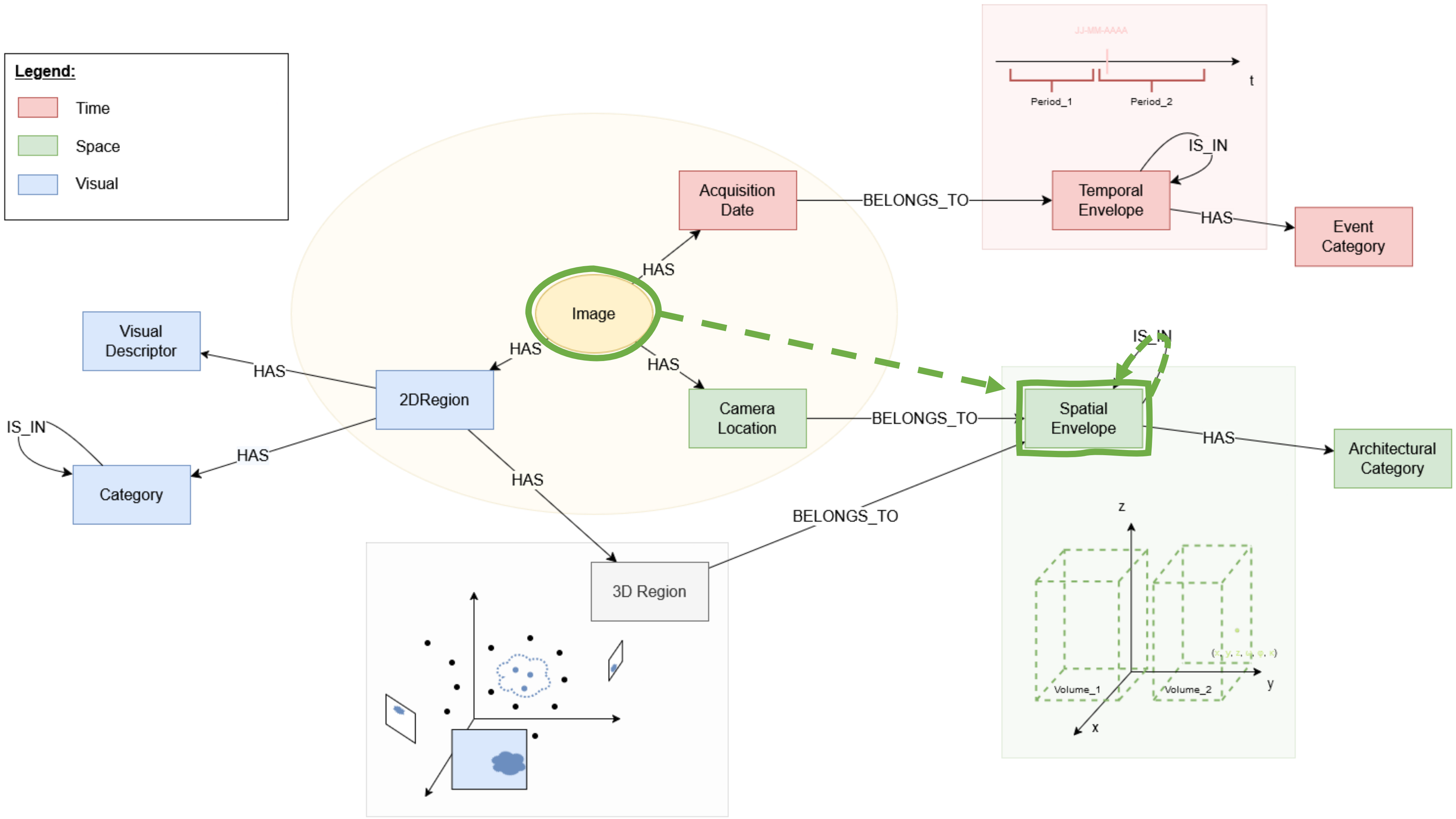


Exploring (I)



Legend:
Time
Space
Visual
Image
HAS
Acquisition Date
BELONGS_TO
Temporal Envelope
IS_IN
HAS
Event Category
t
Period_1
Period_2
HAS
Camera Location
BELONGS_TO
Spatial Envelope
IS_IN
HAS
Architectural Category
HAS
2DRegion
HAS
Visual Descriptor
HAS
Category
IS_IN
HAS
3D Region
BELONGS_TO
z
y
x
Volume_1
Volume_2

Exploring (I)



Legend:
Time
Space
Visual
Acquisition Date
BELONGS_TO
Temporal Envelope
IS_IN
HAS
Event Category
Period_1
Period_2
t
HAS
Image
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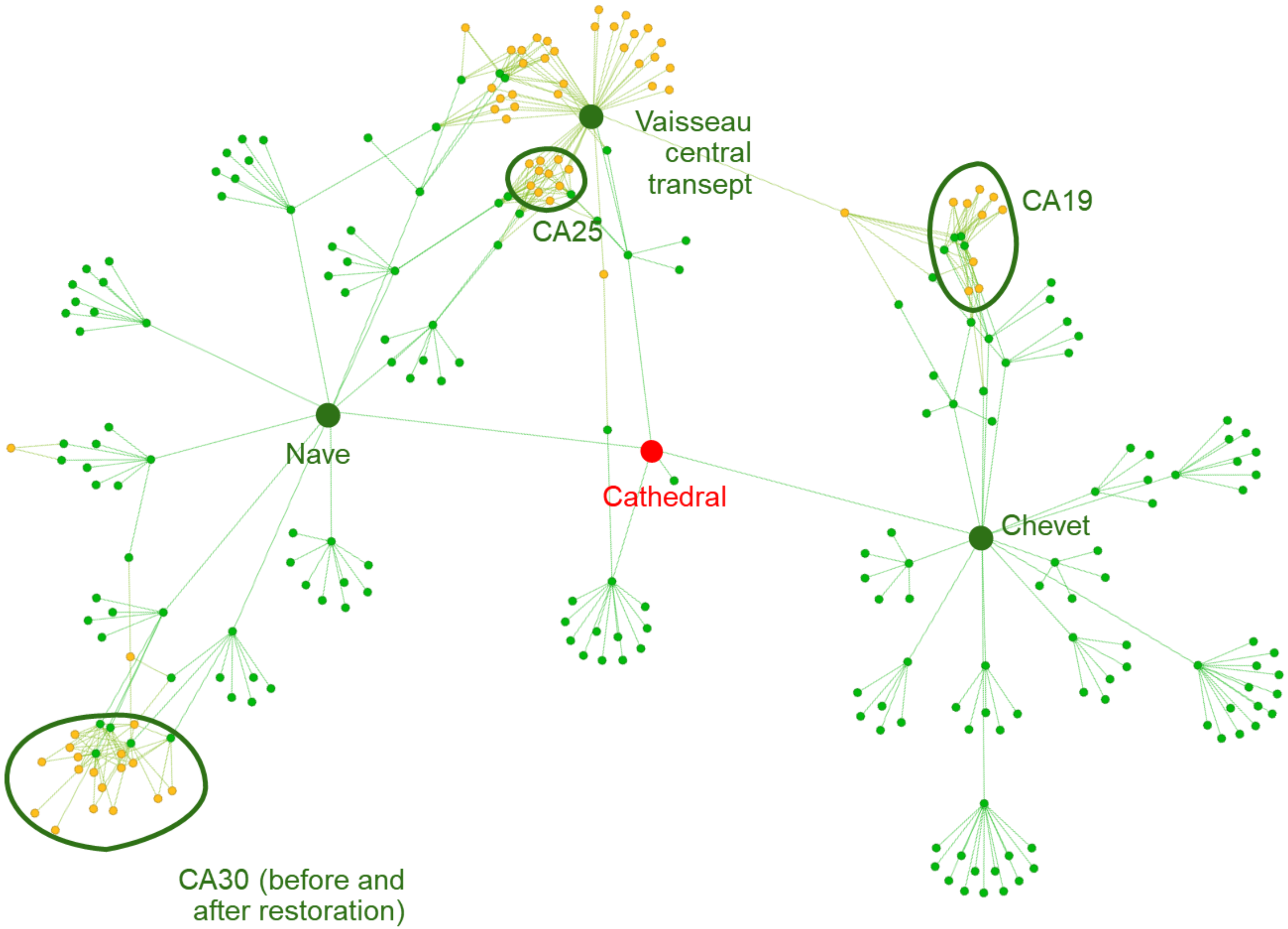
2DRegion
Camera Location
BELONGS_TO
Spatial Envelope
IS_IN
HAS
Architectural Category
IS_IN
HAS
Category
HAS
BELONGS_TO
3D Region
z
y
x
Volume_1
Volume_2

Exploring (I)

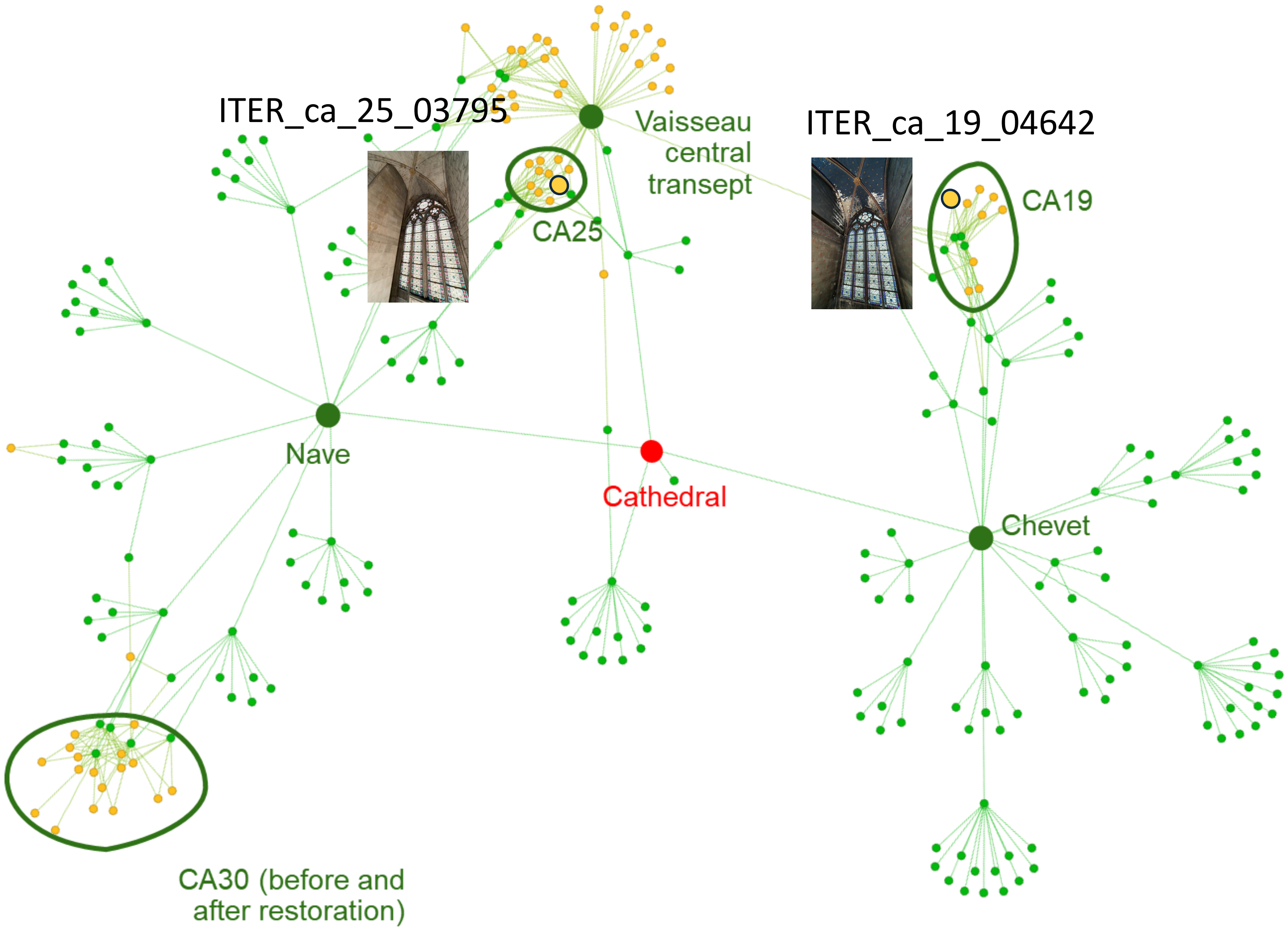
Each Image node is related to {0..n} SpatialEnvelope nodes



Example graph of Images and SpatialEnvelopes

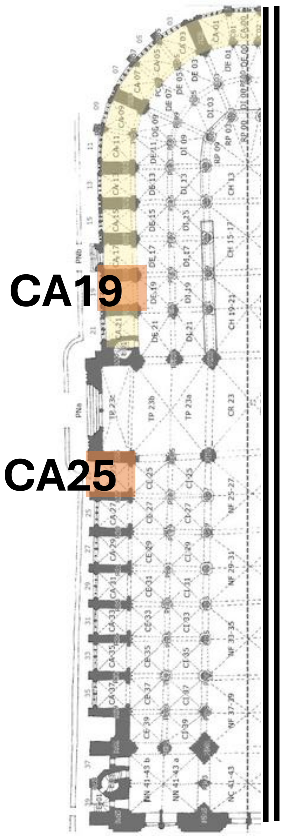
Exploring (I)

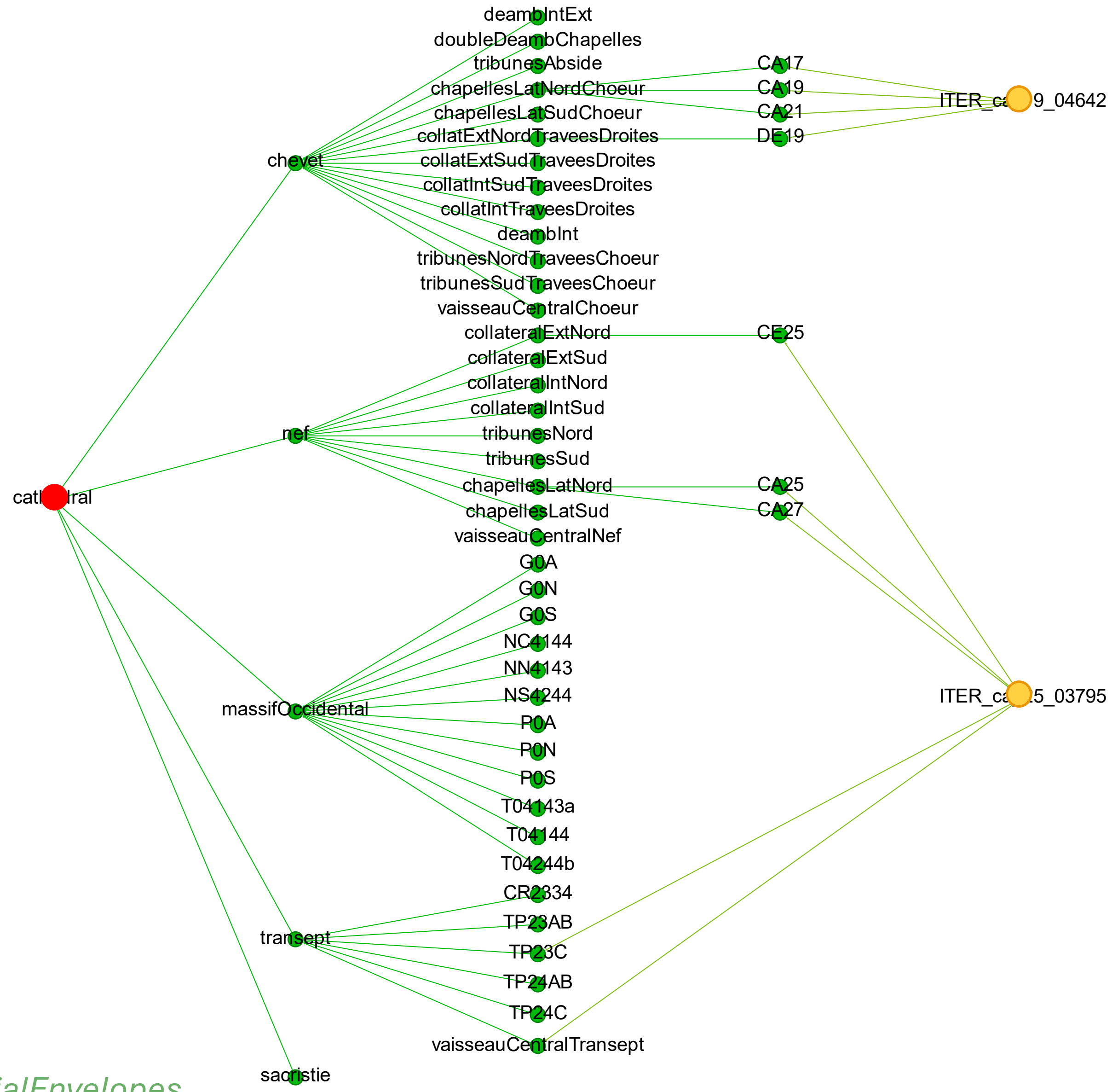
Each Image node is related to {0..n} SpatialEnvelope nodes



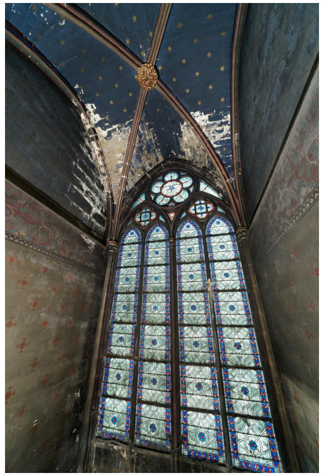
Example graph of Images and SpatialEnvelopes

Exploring (I)

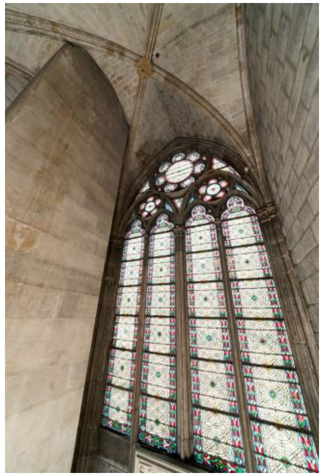




ITER_ca_19_04642

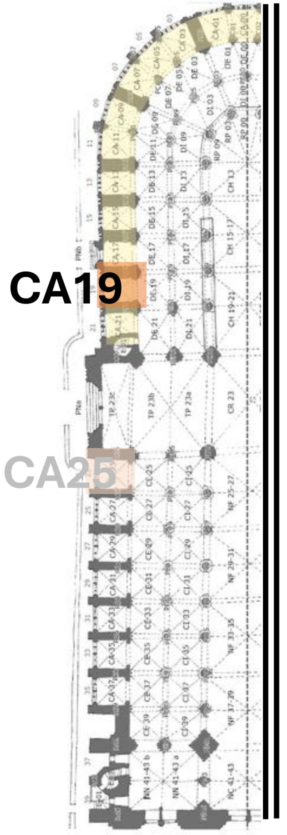


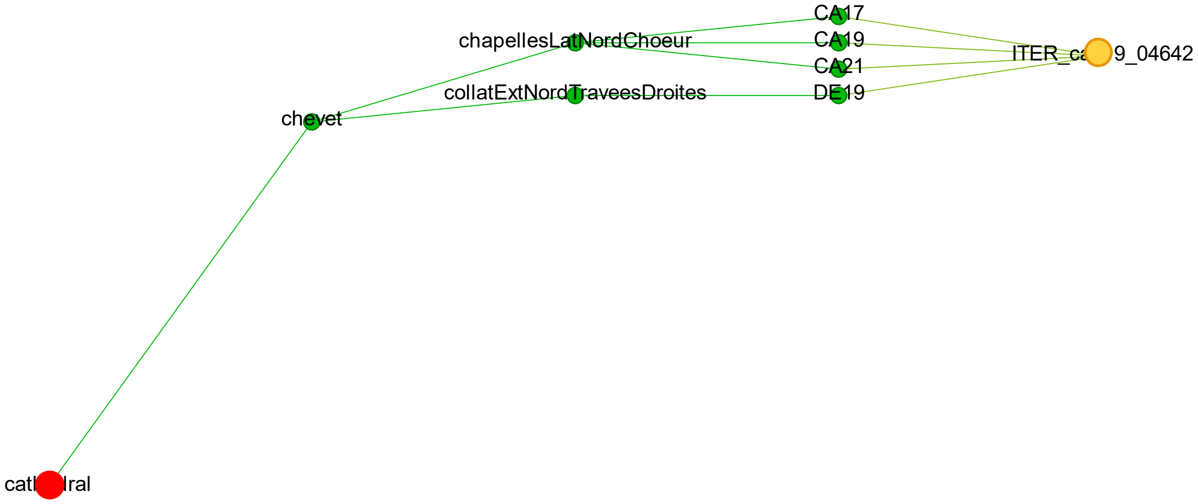
ITER_ca_25_03795



Sub-graph of Images and SpatialEnvelopes of images ITER_ca_19_04642 *and* ITER_ca_25_03795

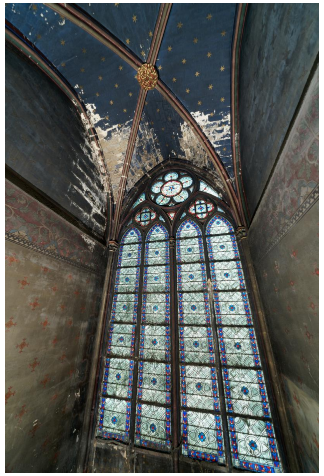
Exploring (I)



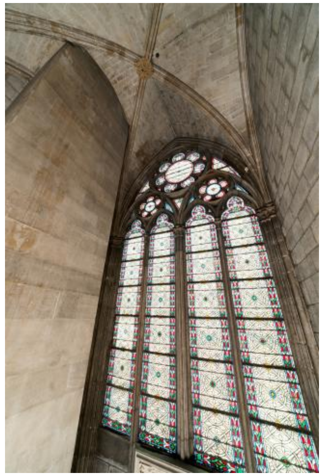


Sub-tree of the SpatialEnvelopes caracterising the image ITER_ca_19_04642

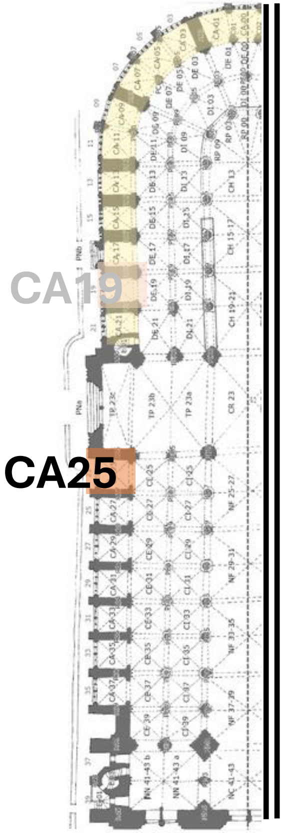
ITER_ca_19_04642



ITER_ca_25_03795

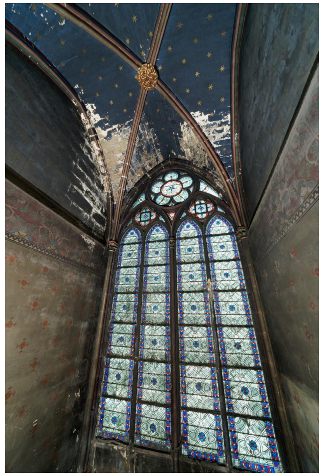


Exploring (I)

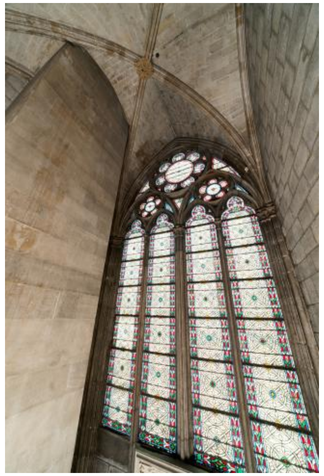


Sub-tree of the SpatialEnvelopes caracterising the image ITER_ca_25_03795

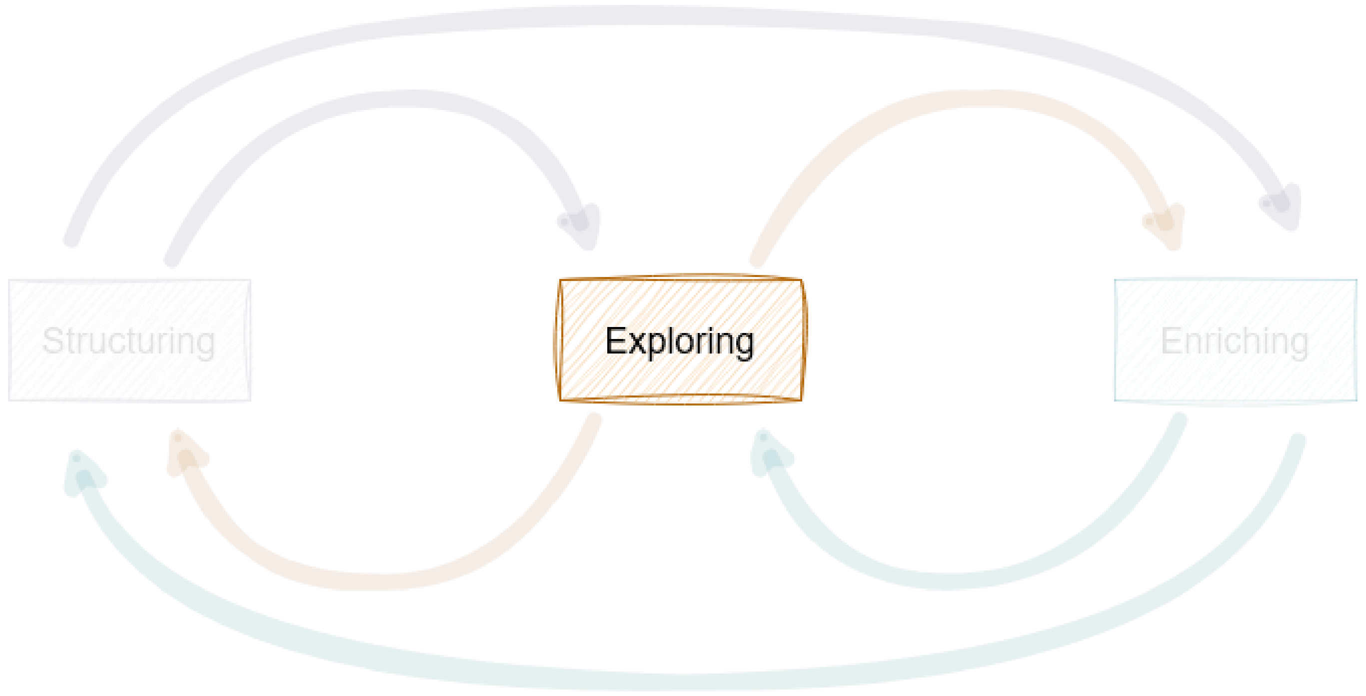
ITER_ca_19_04642



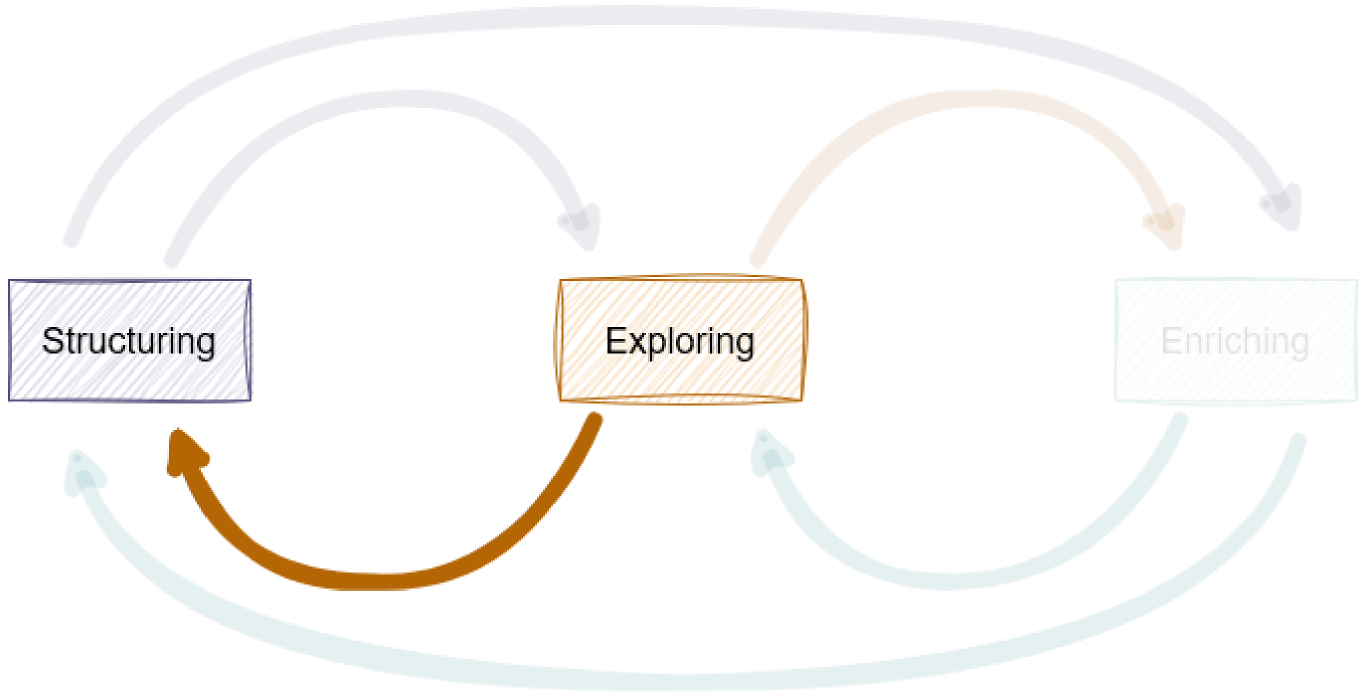
ITER_ca_25_03795



Process

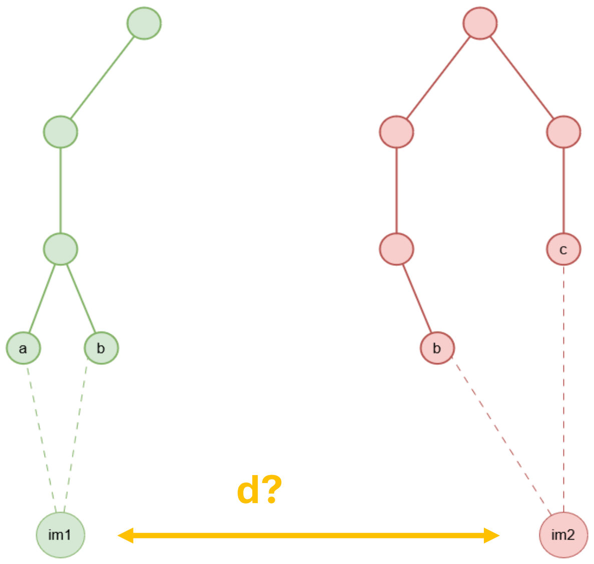


Process



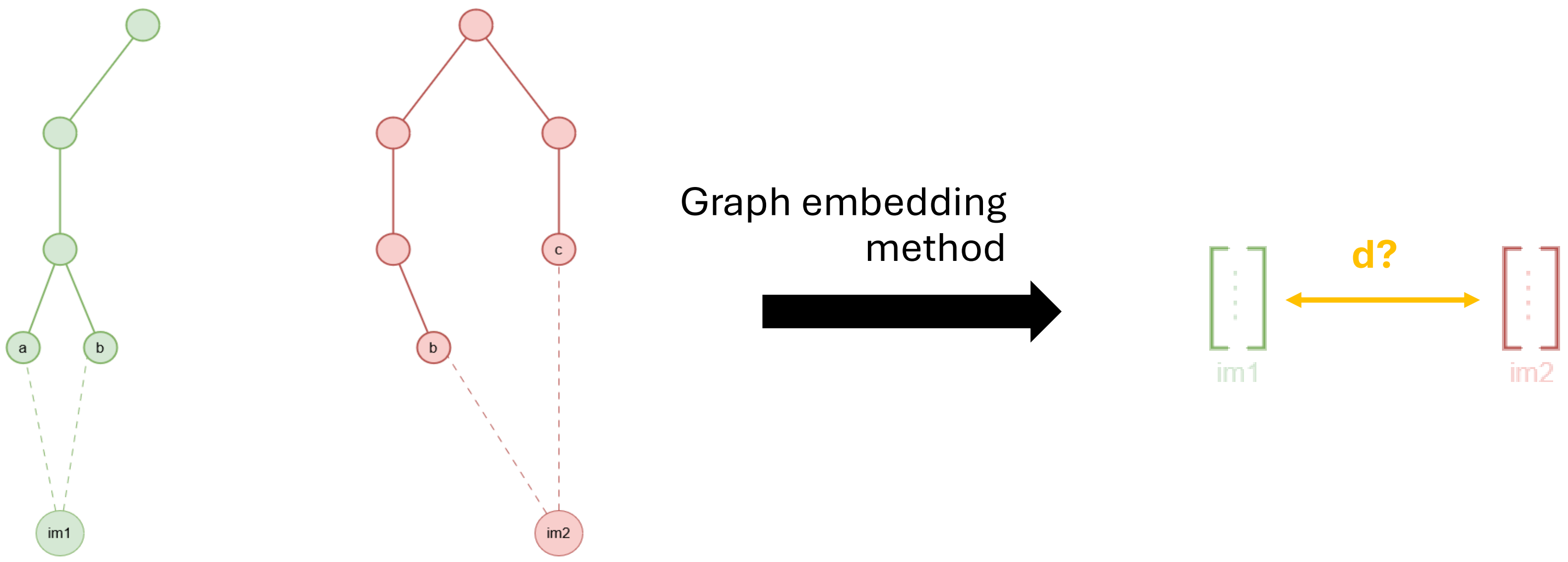
Structuring (II)

- Distances & Data representation



Structuring (II)

- Distances & Data representation



Structuring (II)

- Graph embeddings
  - *Graph embedding converts a graph into a low dimensional space in which the graph information is preserved.* **[Cai et al, 2018]**

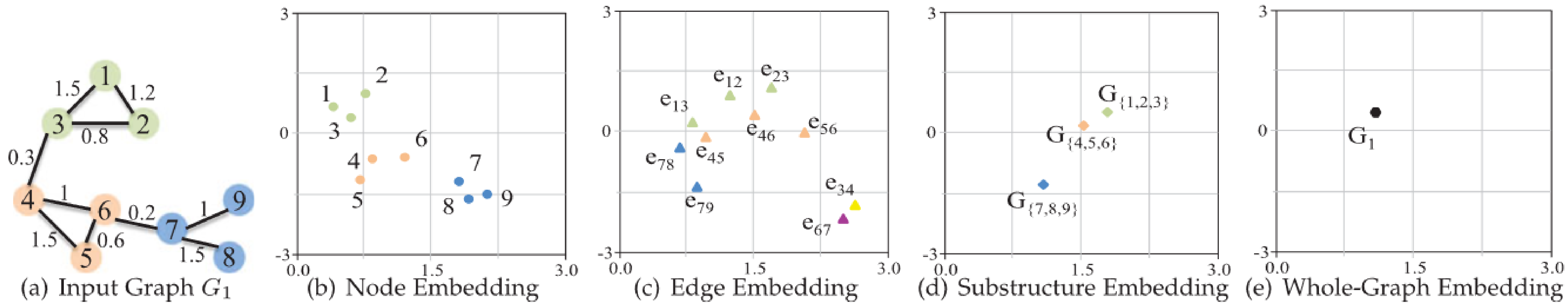


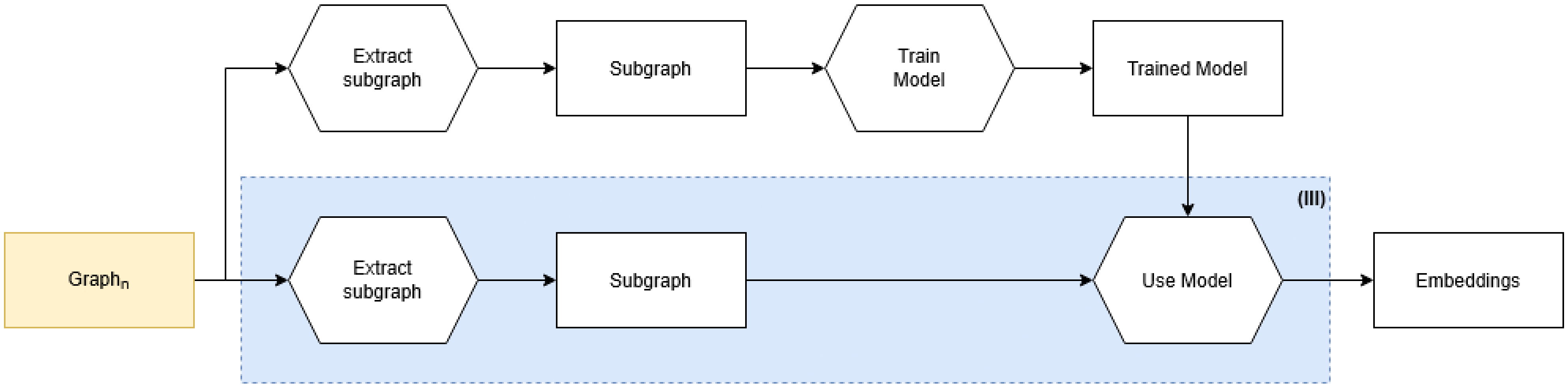
Fig. 1. A toy example of embedding a graph into 2D space with different granularities. $G_{\{1,2,3\}}$ denotes the substructure containing node v_1, v_2, v_3.

Structuring (II)

- Graph embeddings
  - *Graph embedding converts a graph into a low dimensional space in which the graph information is preserved.* **[Cai et al, 2018]**
  - *The challenge [...] is that there is no straightforward way to encode this high-dimensional, non-Euclidean information about graph structure into a feature vector.* **[Hamilton et al, 2018a]**

Structuring (II)

- Graph embeddings
  - Pipeline

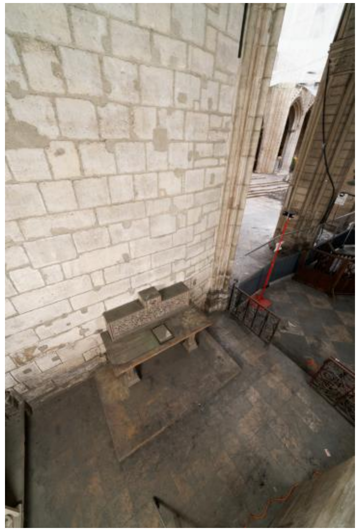


Structuring (II)

- Graph embeddings
    - Pipeline
    - Preliminary Results
        - Images

2

3



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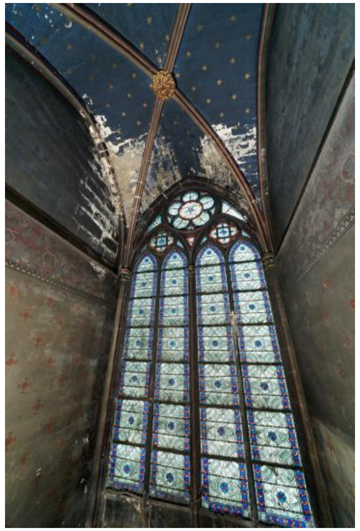
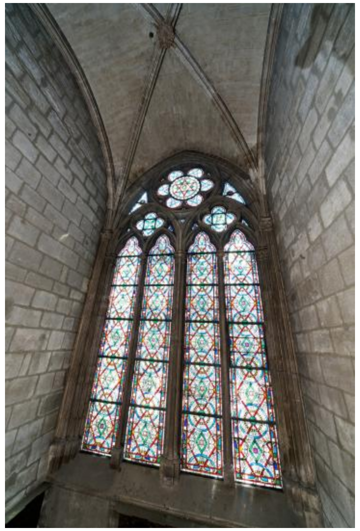


5

6

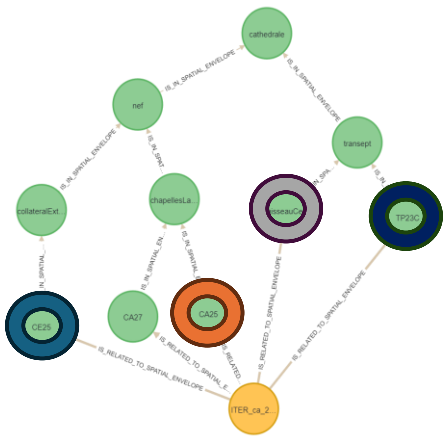
7



Structuring (II)

- Graph embeddings
  - Pipeline
  - Preliminary Results
    - Images
    - SpatialEnvelope sub-trees

2



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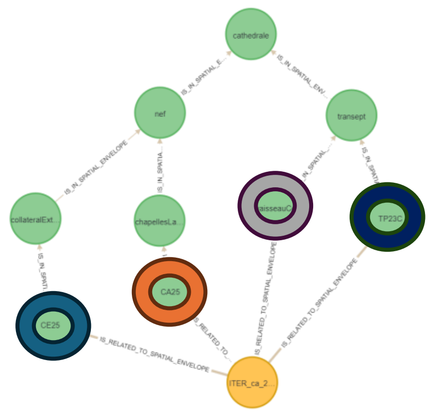

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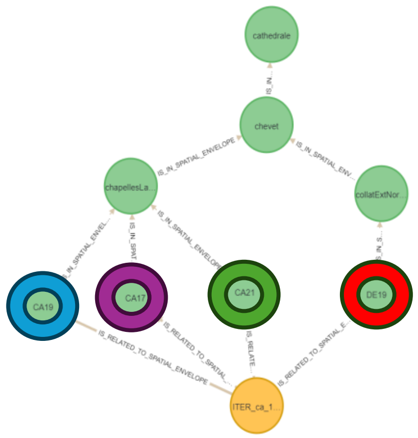

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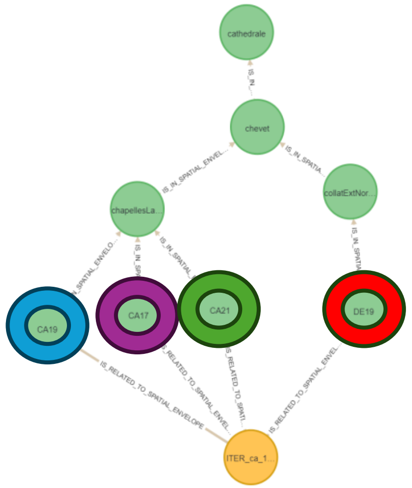

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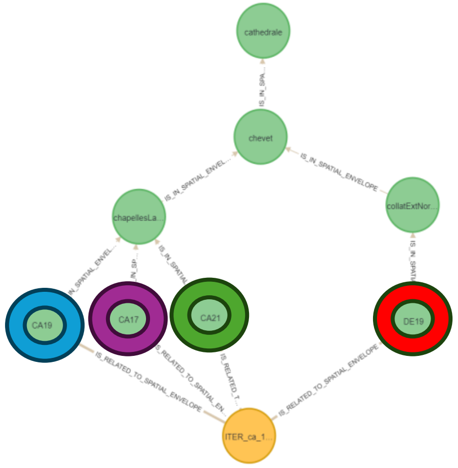

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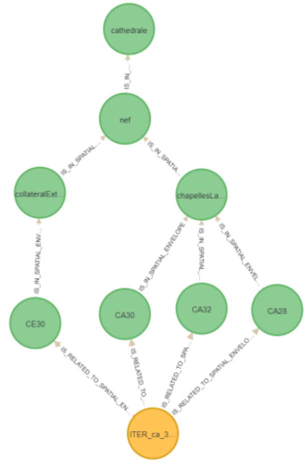

Structuring (II)

- Graph embeddings
  - Pipeline
  - Preliminary Results
    - Images
    - SpatialEnvelope sub-trees
    - Model training

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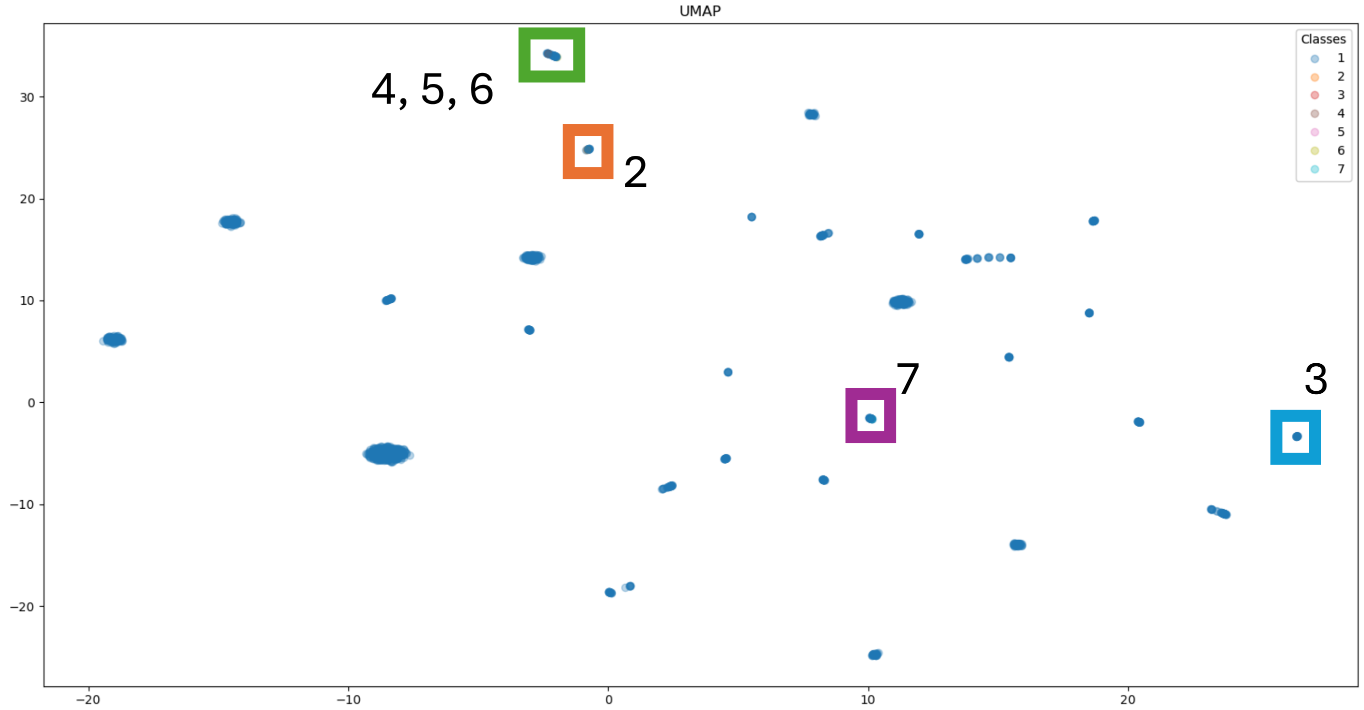
Config file to train the model and generate the node embeddings

Structuring (II)

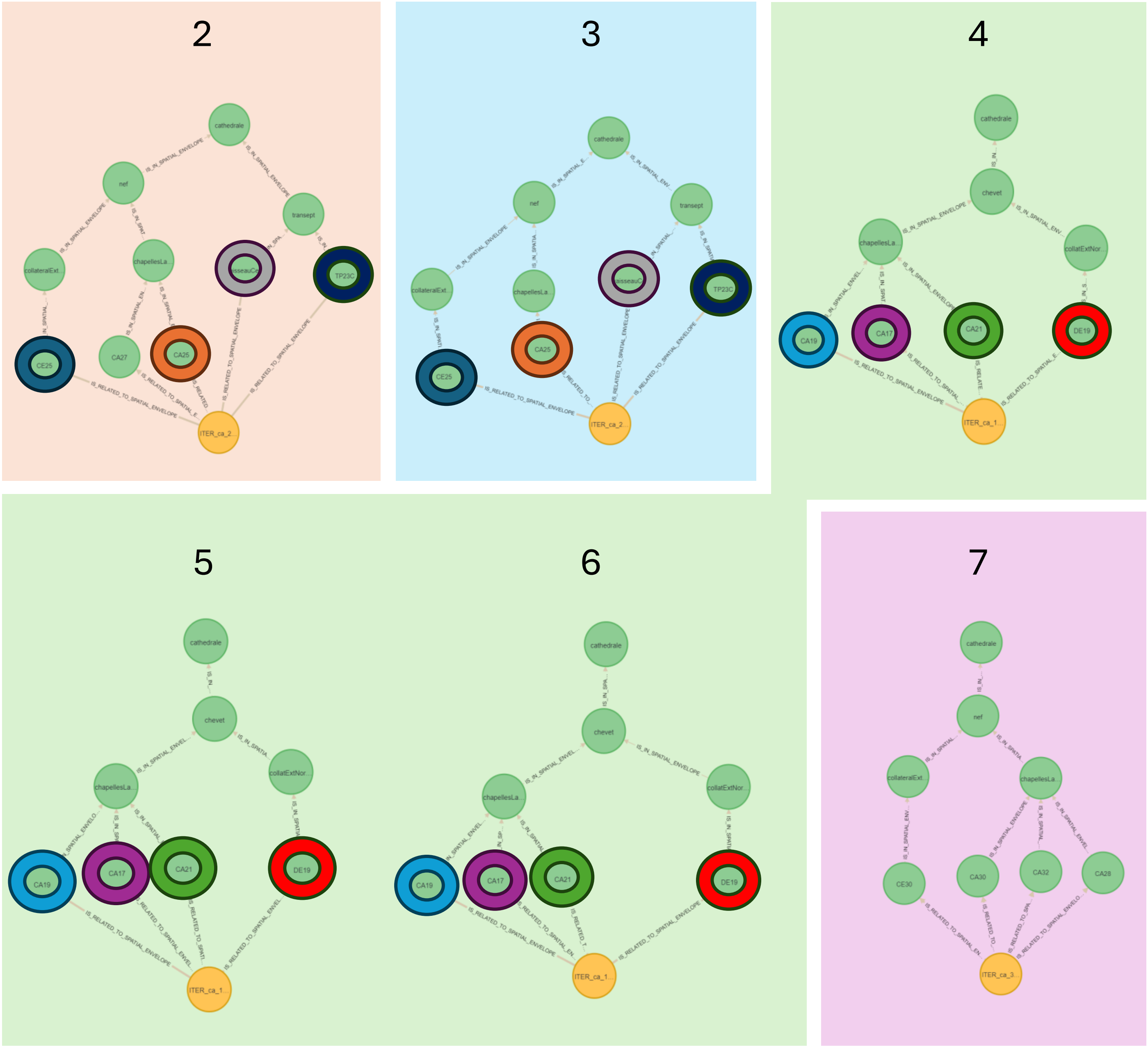
- Graph embeddings
  - Pipeline
  - Preliminary Results
    - Images
    - SpatialEnvelope sub-trees
    - Model training
    - Visualisation



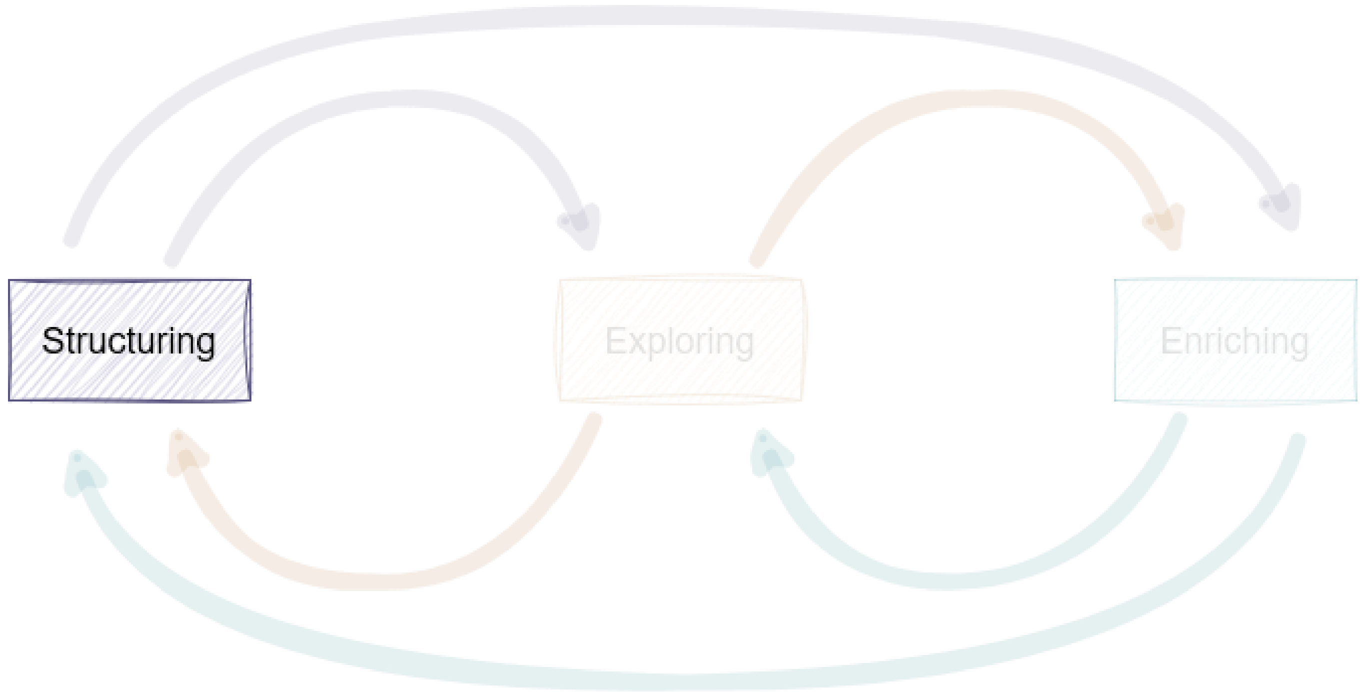
*Using GraphSAGE **[Hamilton et al, 2018b]** and UMAP **[McInnes et al, 2020]***

Structuring (II)

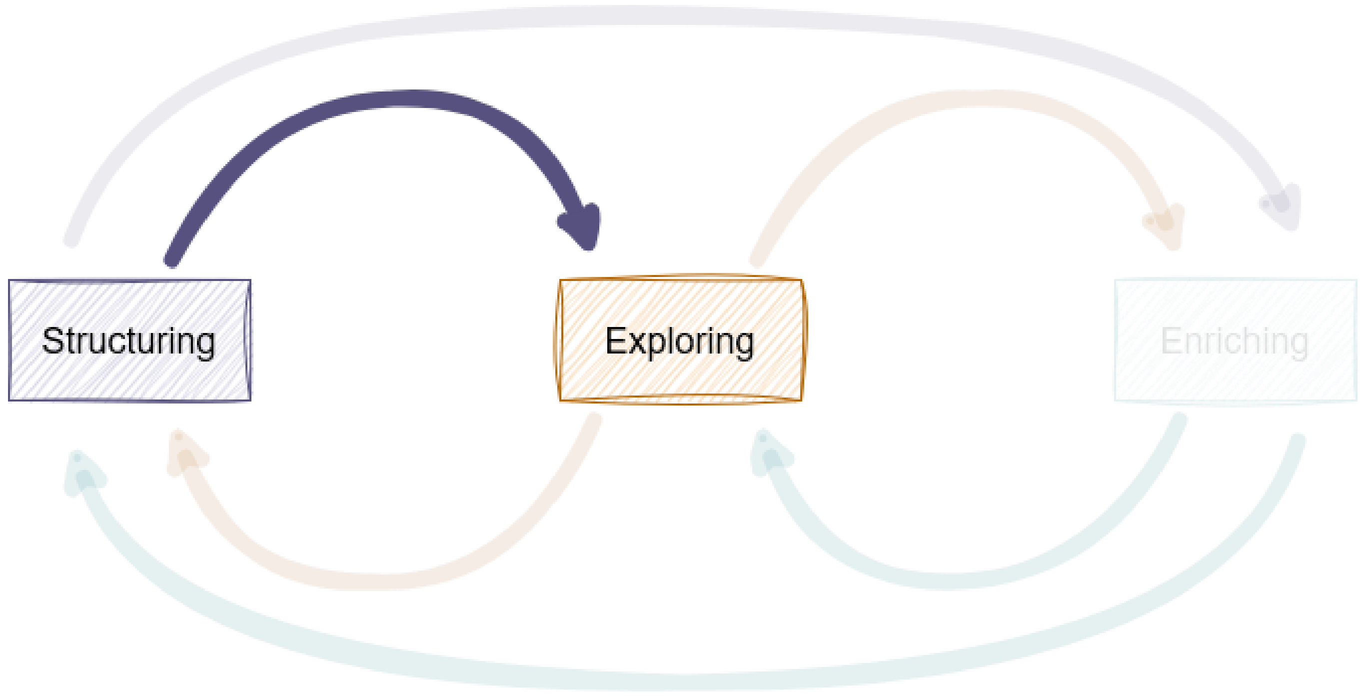
- Graph embeddings
  - Pipeline
  - Preliminary Results
    - Images
    - SpatialEnvelope sub-trees
    - Model training
    - Visualisation
    - Analysis



Process

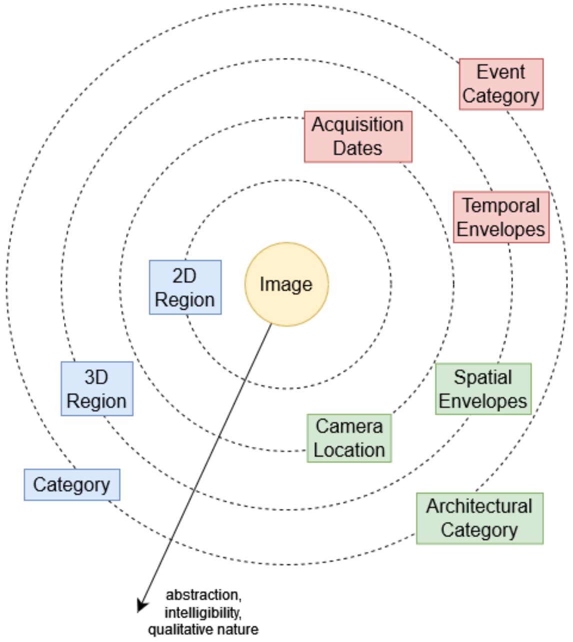


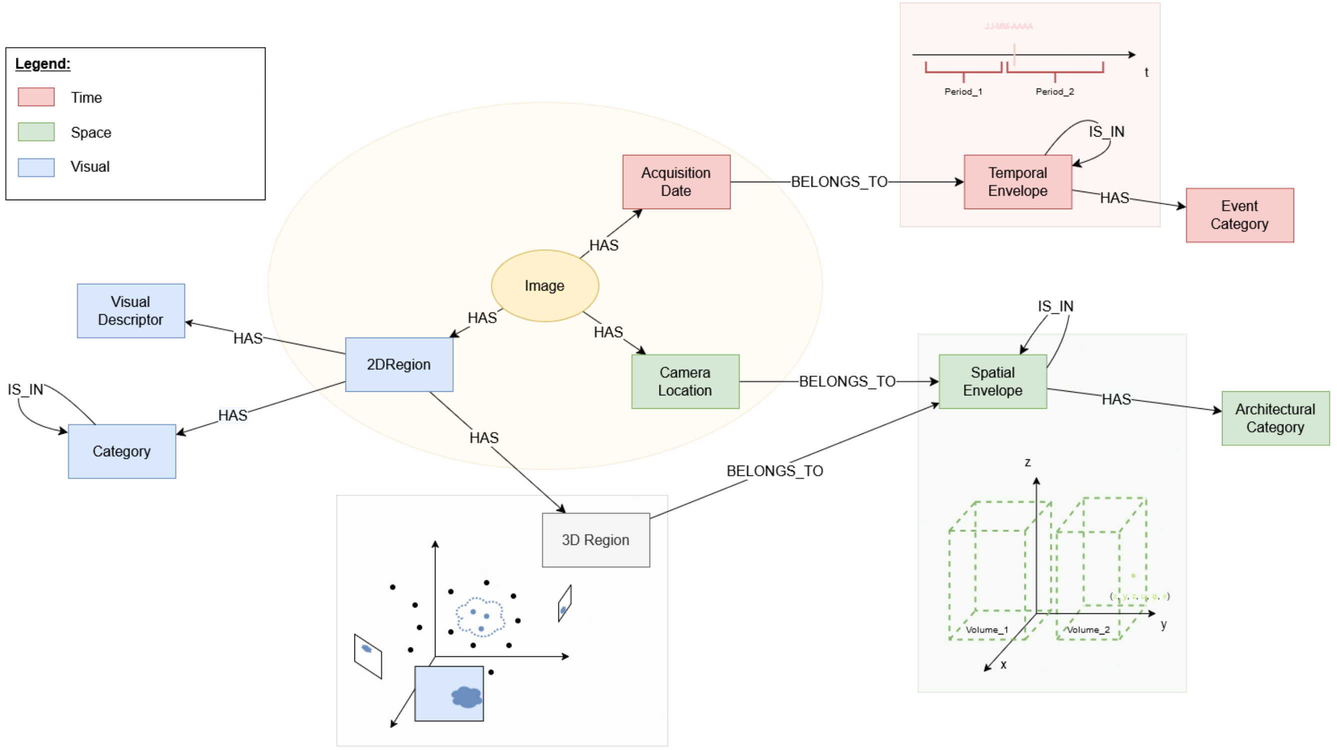
Process



Exploring (II)

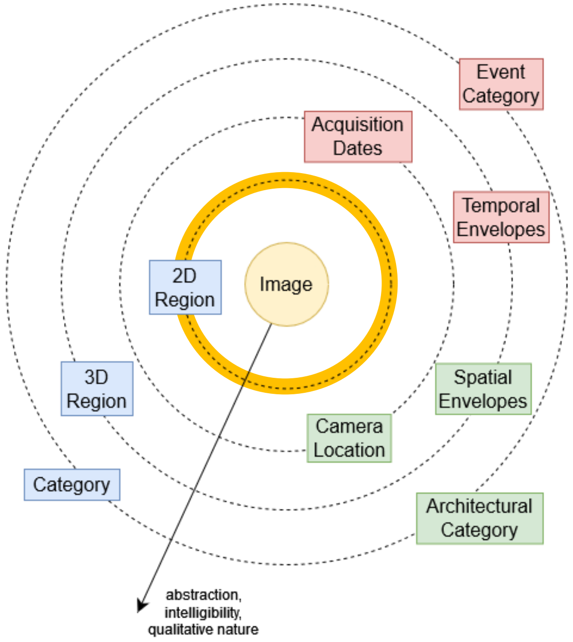
- Exploration based on:
  - Visual descriptors **[Blettery et al, 2023]**

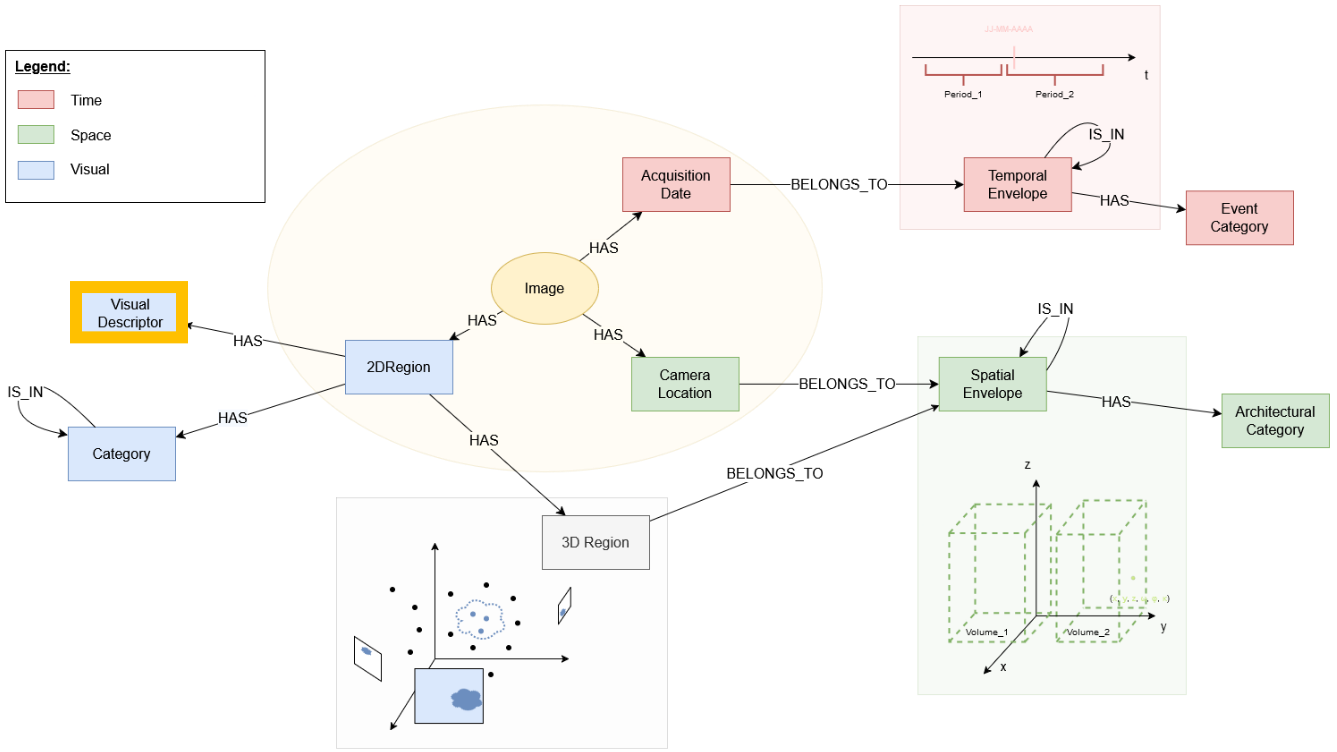




Exploring (II)

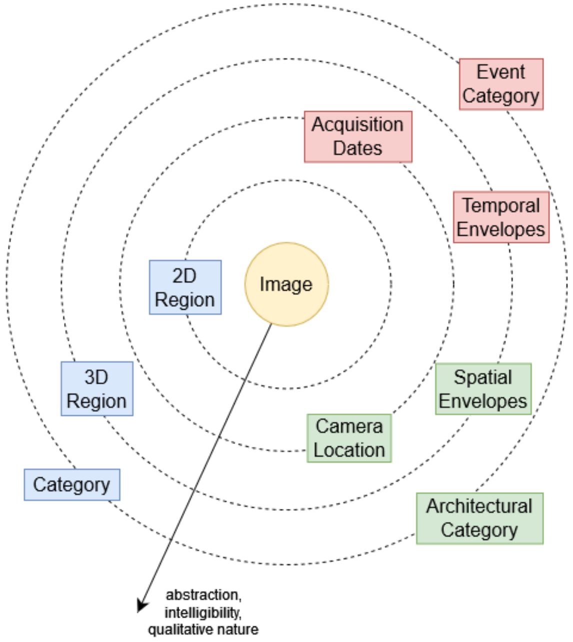
- Exploration based on:
  - Visual descriptors **[Blettery et al, 2023]**

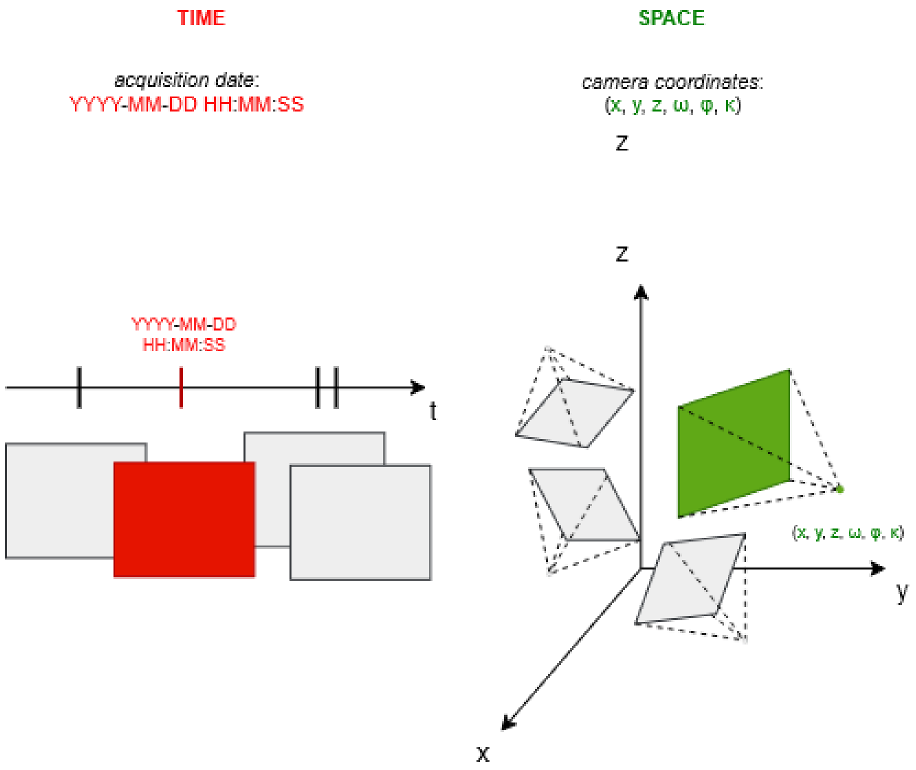




Exploring (II)

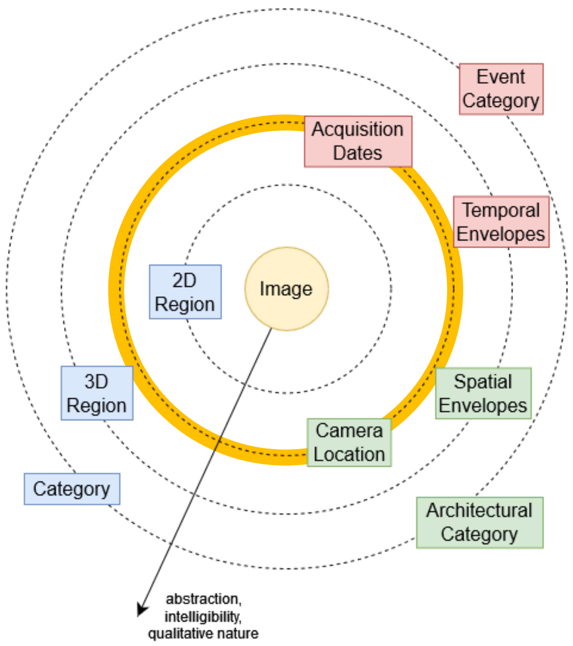
- Exploration based on:
  - Visual descriptors
  - Data values

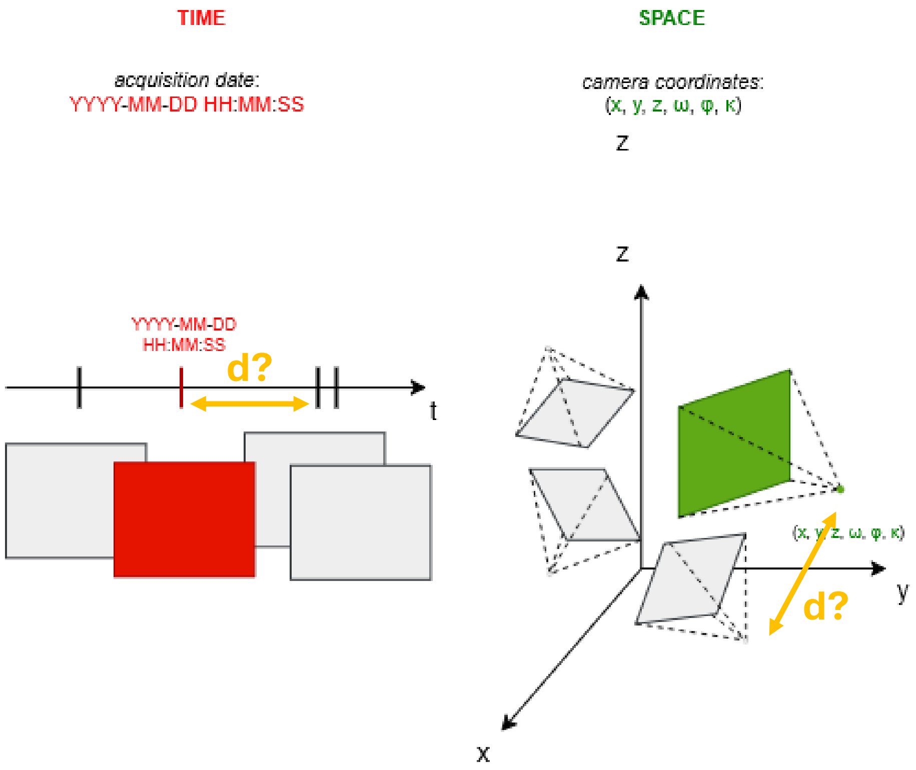




Exploring (II)

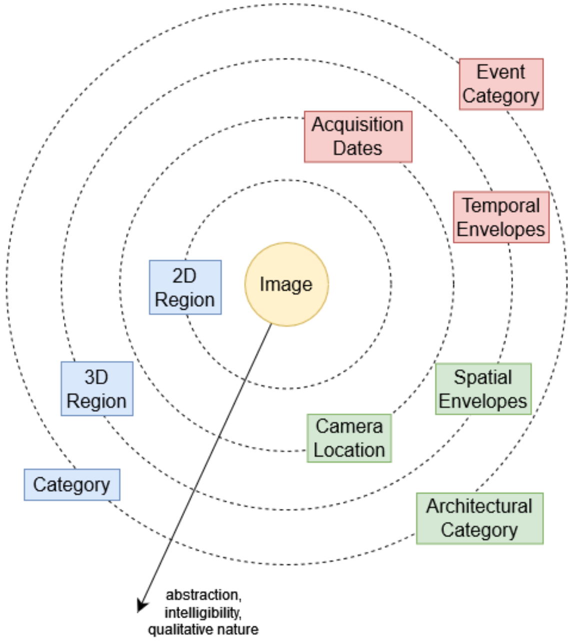
- Exploration based on:
  - Visual descriptors
  - Data values

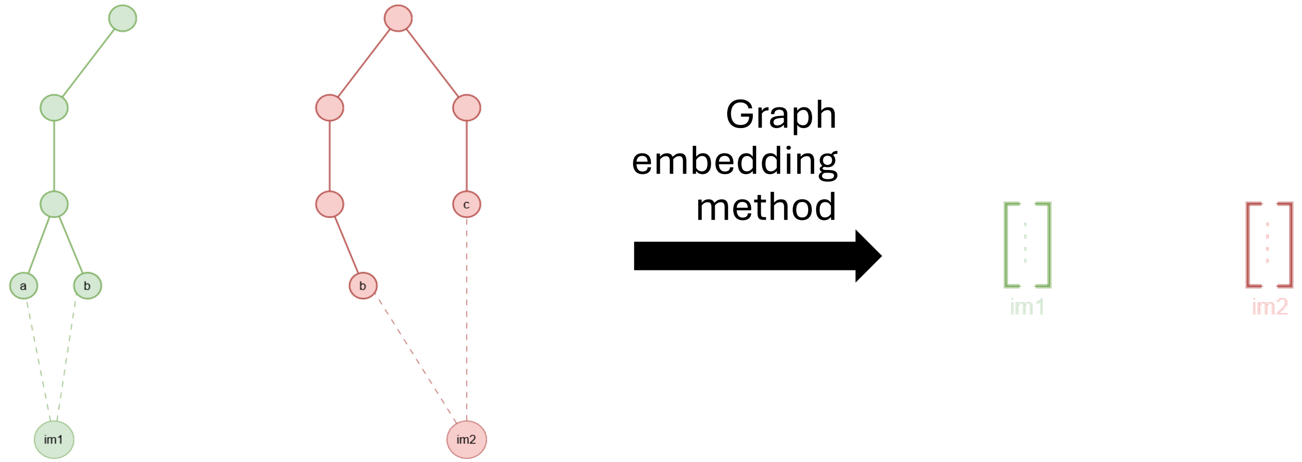




Exploring (II)

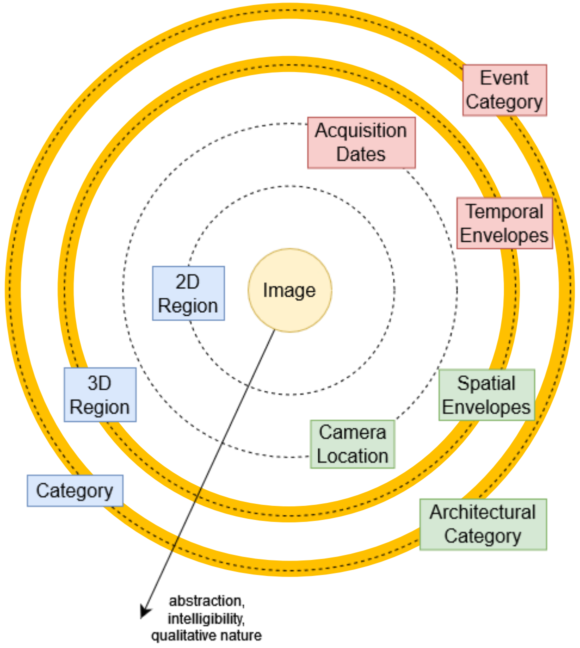
- Exploration based on:
  - Visual descriptors
  - Data values
  - Data structures (embeddings)

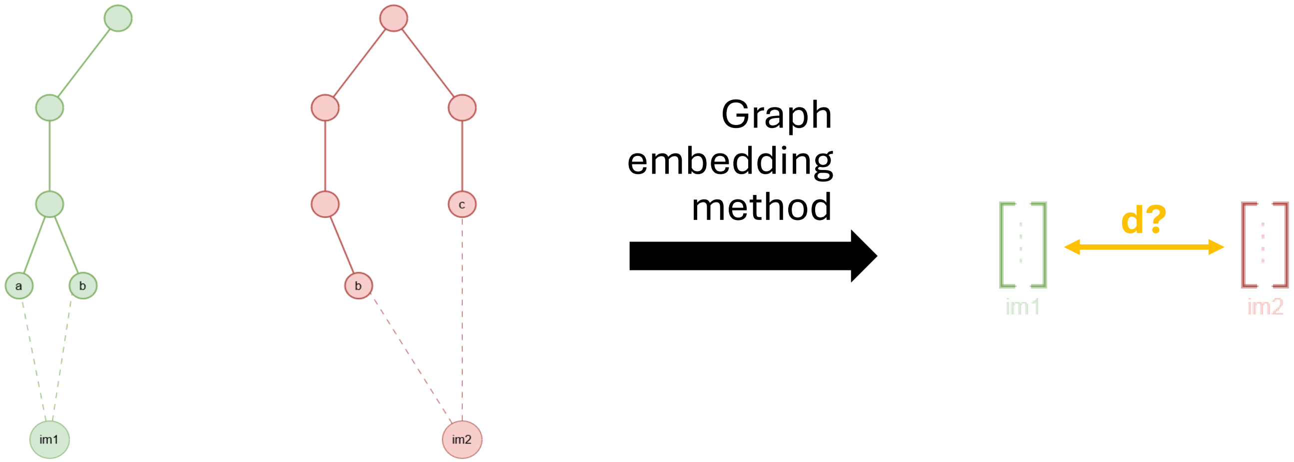




Exploring (II)

- Exploration based on:
  - Visual descriptors
  - Data values
  - Data structures (embeddings)





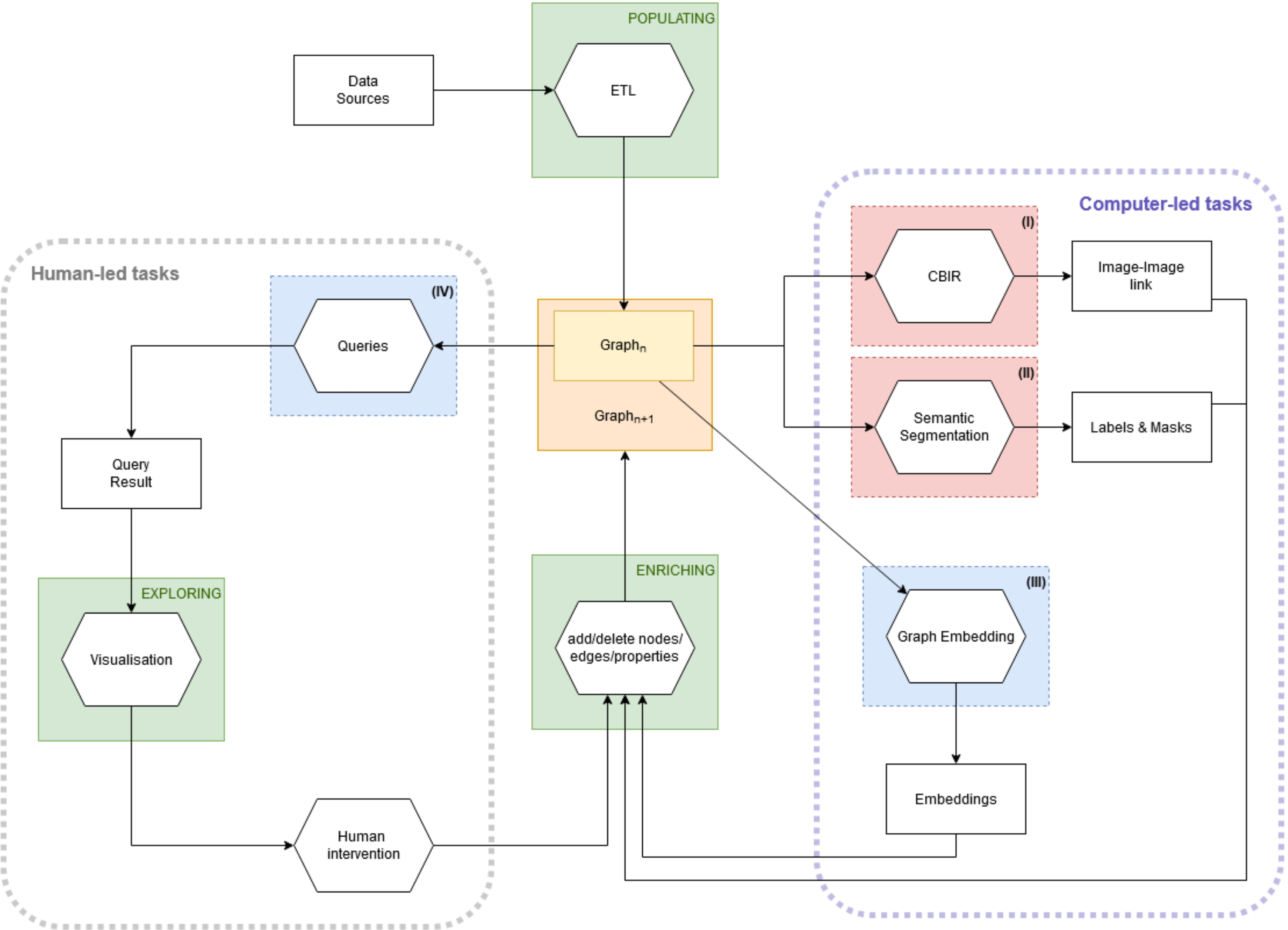
Exploring (II)

- Exploration techniques:
  - Clustering
  - Nearest Neighbours
  - Dimensionality Reduction
  - …

Conclusion

- Global approach
  - Human-led tasks
  - Computer-led tasks

- Other tasks
  - Semantic Segmentation **[Réby et al, 2023]**
  - Content-based Image Retrieval **[Blettery et al, 2023]**



Bibliography

- **[Blettery et al, 2023]** Emile Blettery, Valérie Gouet-Brunet. Re-ranking Image Retrieval in Challenging Geographical Iconographic Heritage Collections. 20th International Conference on Content-Based Multimedia Indexing, Sep 2023, Orléans, France. https://dl.acm.org/doi/10.1145/3617233.3617259.
- **[Cai et al, 2018]** HongYun Cai, Vincent W. Zheng, et Kevin Chen-Chuan Chang, « A Comprehensive Survey of Graph Embedding: Problems, Techniques, and Applications », *IEEE Transactions on Knowledge and Data Engineering* 30, nº 9 (septembre 2018): 1616-37, https://doi.org/10.1109/TKDE.2018.2807452.
- **[Hamilton et al, 2018a]** William L. Hamilton, Rex Ying, et Jure Leskovec, « Representation Learning on Graphs: Methods and Applications » (arXiv, 10 avril 2018), http://arxiv.org/abs/1709.05584.
- **[Hamilton et al, 2018b]** William L. Hamilton, Rex Ying, et Jure Leskovec, « Inductive Representation Learning on Large Graphs » (arXiv, 10 septembre 2018), http://arxiv.org/abs/1706.02216.
- **[Hamilton et al, 2020]** William L Hamilton, « Graph Representation Learning », Synthesis Lectures on Artificial Intelligence and Machine Learning 3, août 2020.
- **[McInnes et al, 2020]** Leland McInnes, John Healy, et James Melville, « UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction » (arXiv, 17 septembre 2020), https://doi.org/10.48550/arXiv.1802.03426.
- **[Réby et al, 2023]** Kévin Réby, Anaïs Guillem, Livio De Luca. Semantic Segmentation using Foundation Models for Cultural Heritage: an Experimental Study on Notre-Dame de Paris. 4th ICCV Workshop on Electronic Cultural Heritage, Computer Vision Foundation, Oct 2023, Paris, France. https://openaccess.thecvf.com/content/ICCV2023W/e-Heritage/html/Reby_Semantic_Segmentation_Using_Foundation_Models_for_Cultural_Heritage_an_Experimental_ICCVW_2023_paper.html.

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Thank you for your attention!